Technical Supplement: Honest Tensor-Functional Inference

Revised September 13, 2026

This supplement contains the complete supplied proofs supporting the accompanying revised main paper. The main paper states the final central results and empirical findings; this document records the auxiliary comparisons, calibrations, computational contracts and proofs. Fixed-budget intermediate results are retained here as proof ingredients, not as unresolved final rate claims. All formal results have internal status PROVED; this means a checked supplied derivation, not formal verification or external peer review.

The notation is local to each explicit experiment: D_{amb} is ambient dimension, d polynomial degree, C_{diag} a diagonal-envelope coefficient, ε_A an approximation allowance and η_k a pilot operator bound. The main paper's $q_K = 2/K$ is denoted q in scalar-threshold proofs; its z-quantile notation (or z_{val}) is used for Gaussian critical values. Canonical complexity J and support overlap H are distinguished.

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1 Experiment and parameter classes

Let $K \geq 2$, $p_k \geq 1$, and $D_{\text{amb}} = \prod_{k=1}^{K} p_k$. We observe

$$Z = A + \tau G \in \mathbb{R}^{p_1 \times \cdots \times p_K}, \qquad \operatorname{vec}(G) \sim N(0, I_{D_{\text{amb}}}), \qquad \tau > 0. \tag{1}$$

The noise level is known in the baseline model. Section 5 states different, explicitly calibrated noise and rank assumptions. Equivalently, Z is the sufficient sample mean of independent full observations $W_j = A + \sigma G_j$, $j = 1, \ldots, n$, with known σ and $\tau = \sigma/\sqrt{n}$. This is an exact Gaussian mean equivalence, not an equivalence to scalar-response tensor regression. The observed variables are the tensors W_j, or Z; A is a fixed unknown parameter. Factor and core representations are nonunique coordinates, not additional latent random variables.

Write $A_{(k)}$ for the mode-k unfolding and $A_{\langle k \rangle}$ for the matrix joining modes $1, \ldots, k$ against $k+1, \ldots, K$. For supplied positive integer caps, define

$$\Theta_{\text{Tu}}(M, \bar{r}) = \{A : \|A\|_F \leq M,\ \operatorname{rank}(A_{(k)}) \leq \bar{r}_k,\ k = 1, \ldots, K\}, \tag{2}$$

$$\Theta_{\text{TT}}(M, \bar{r}) = \{A : \|A\|_F \leq M,\ \operatorname{rank}(A_{\langle k \rangle}) \leq \bar{r}_k,\ k = 1, \ldots, K-1\}. \tag{3}$$

The known bound M is positive. Actual ranks may be smaller, including zero; nonzero singular values may be arbitrarily small. No incoherence or coordinate sparsity condition is imposed. The main experiment assumes Gaussian isotropic noise and exact caps. Appendix C states separately calibrated noise and approximation extensions; missing observations remain untreated.

The prescribed loading has the form

$$L = \sum_{i=1}^{m} w_i V_i, \qquad V_i = \bigotimes_{k=1}^{K} v_{ki}, \qquad v_{ki}^{\top} v_{kj} = \mathbf{1}\{i = j\}, \qquad \|w\|_2 = 1. \tag{4}$$

Here $m \le \min_k p_k$, and each loading family is supplied independently of the data. Thus $\|L\|_F = 1$. Although this is restrictive, L need not be sparse in the observation basis: known orthogonal mode rotations can make it dense. The estimand and relevant observations are

$$\psi(A) = \langle L, A\rangle = \sum_i w_i a_i, \qquad a_i = \langle A, V_i\rangle, \qquad d_i = \langle Z, V_i\rangle = a_i + \tau g_i, \tag{5}$$

where the g_i are independent standard normals.

An interval $C(Z)$ is honest at level $1-\alpha$ on Θ if

$$\inf_{A\in\Theta} \mathbb{P}_A\{\psi(A) \in C(Z)\} \ge 1-\alpha. \tag{6}$$

This is a finite-sample requirement, not a pointwise limit along a fixed-rank stratum. Length and expected length are the primary criteria. For example,

$$\mathcal{L}_0(\Theta) = \inf_{C:\,(6)} \mathbb{E}_0|C|, \qquad \mathcal{L}_{\max}(\Theta) = \inf_{C:\,(6)} \sup_{A\in\Theta} \mathbb{E}_A|C|. \tag{7}$$

The infima range over all measurable intervals, allowing independent randomization. The usual squared Frobenius estimation loss has population risk $R_A(B) = \|B-A\|_F^2 + D_{\mathrm{amb}}\sigma^2$ and empirical risk $R_n(B) = n^{-1}\sum_j \|W_j - B\|_F^2$. We do not minimize these risks or require a full-tensor estimator.

The fixed-budget benchmark comparison in experiment (1) uses $p_k = m = p$, fixed $K \ge 3$, fixed rank caps, and $M = \tau\sqrt{p}$. A concrete sequence is $M = \sigma = 1$, $n = p$, so $D_{\mathrm{amb}} = n^K$. Section 4 removes the relation between M and τ for the sharp full-data results and extends them to flat orthogonal loadings in unequal ambient dimensions.

The structural and probabilistic assumptions have distinct roles, summarized in Table 1. A1–A4 define the inferential problem; A5 is used for the original asymptotic comparisons. The norm/noise relation in A5 is not required in Section 4.

	Assumption	Role and restriction
A1	Full observation with independent Gaussian coordinates and known τ	Gives exact diagonal noise, direct calibration and Gaussian likelihood identities.
A2	Supplied M and Tucker or TT upper rank bounds	Defines the outer honesty class; actual rank and signal strength are unrestricted below it.
A3	Prescribed loading in (4)	Gives orthogonal noise coordinates and the diagonal envelope; data-selected loadings are excluded; arbitrary prescribed loadings are treated separately in Section 5.
A4	Finite-sample uniform coverage in (6)	Coverage includes zero and local sequences approaching lower-rank strata.
A5	Equal dimensions, fixed $K \ge 3$, fixed caps and $M = \tau\sqrt{p}$	Produces the stated vanishing length ratio; not needed for the finite-sample interval.

Table 1: Assumptions, their purpose and the scope they exclude. Known upper bounds are not replaced by same-data estimates anywhere in the stated procedure.

2 Interval construction

Define

$$q = \frac{2}{K}, \quad J_{\mathrm{Tu}} = \min_k \prod_{\ell\neq k} \bar{r}_\ell, \quad J_{\mathrm{TT}} = \prod_{k=1}^{K-1} \bar{r}_k, \quad B = J^{1-q/2} M^q, \tag{8}$$

where J is chosen for the applicable class. Set

$$\lambda = \tau\sqrt{2\log(2m/\alpha)}, \qquad \widehat{\psi}_{\mathrm{NL}} = \sum_i w_i \,\mathrm{soft}_\lambda(d_i), \qquad \mathrm{soft}_\lambda(x) = \mathrm{sign}(x)(|x| - \lambda)_+. \tag{9}$$

The three candidate intervals have radii

$$h_{\mathrm{NL}} = \|w\|_\infty B(2\lambda)^{1-q}, \qquad I_{\mathrm{NL}} = [\widehat{\psi}_{\mathrm{NL}} - h_{\mathrm{NL}}, \widehat{\psi}_{\mathrm{NL}} + h_{\mathrm{NL}}], \tag{10}$$

$$h_G = z_{1-\alpha/2}\tau, \qquad I_G = [\langle L, Z\rangle - h_G, \langle L, Z\rangle + h_G], \tag{11}$$

$$h_0 = M\left(\sum_{i=1}^{s} w_{(i)}^2\right)^{1/2}, \qquad I_0 = [-h_0, h_0], \tag{12}$$

where $|w_{(1)}| \geq \cdots \geq |w_{(m)}|$ and s is the smaller of m and the smallest supplied rank cap. Choose

$$C_* = I_{j_*}, \qquad j_* \in \arg\min_{j \in \{\mathrm{NL}, G, 0\}} h_j, \tag{13}$$

with a fixed rule for ties. This choice depends only on known quantities. Choosing an interval after examining its center or an estimated rank is a different procedure and is not justified by the proof.

The constants in (8) calibrate the threshold procedure (9); the direct alternative (11) supplies a valid fallback. Only m contractions are used. For coordinate loadings, computation reads m tensor entries and takes $O(m)$ operations. Obtaining contractions for a dense rotated loading has additional cost, which can be large. The procedure does not estimate a Tucker core, TT cores, factors or actual ranks.

3 Statistical theory

All numbered theorems and propositions are **PROVED**, with complete derivations in the appendix. The result-status table separates the proved statements from open computational and adaptive questions. The results use the experiment and classes of Section 1. Lower-bound constants are stated for $0 < \alpha < 1/4$ when this simplifies positivity.

3.1 Diagonal geometry and honest coverage

Theorem 1 (Orthogonal diagonal compressibility)**.** *For either class in* (2)–(3)*, under loading* (4)*,*

$$\sum_{i=1}^{m} |\langle A, V_i\rangle|^{2/K} \leq J^{1-1/K} M^{2/K}. \tag{14}$$

No minimum singular value is required.

The proof expands a canonical tensor representation into at most J product terms whose squared component weights sum to $\|A\|_F^2$. For TT tensors, these terms need not be orthogonal. A prefix-weight identity establishes the sum exactly; assuming termwise orthogonality would be an incorrect shortcut. Hölder's inequality and Bessel's inequality then give (14). The rank factor may be loose, particularly for growing caps.

For rank cap one the diagonal image is exactly the ball $\{a : \sum_i |a_i|^{2/K} \leq M^{2/K}\}$, as shown in Lemma 2. Thus the diagonal-only statistical experiment is precisely a classical Gaussian sequence problem on an ℓ_q ball. The full tensor experiment contains additional off-diagonal information. This identity permits a sharp benchmark for the restricted observation experiment; it does not remove the additional information in the rest of the tensor.

Theorem 2 (Finite-sample honest interval)**.** *For every $0 < \alpha < 1$, each interval in (10)–(12) is honest on the applicable cap class. Consequently, C_* is honest and has deterministic length*

$$|C_*| = 2\min\{h_{\mathrm{NL}}, h_G, h_0\}. \tag{15}$$

The event $\max_i |\tau g_i| \le \lambda$ has probability at least $1-\alpha$. On this event soft thresholding satisfies

$$|\operatorname{soft}_\lambda(a_i + \tau g_i) - a_i| \le \min(|a_i|, 2\lambda) \le |a_i|^q (2\lambda)^{1-q}.$$

Summing proves the nonlinear interval bound. No central limit theorem, rank recovery or pilot remainder is needed. Known M is essential to this deterministic calibration.

3.2 Two lower-bound comparisons

For the remainder of this subsection let all dimensions equal p, let $E_i = e_i^{\otimes K}$, and put

$$L = p^{-1/2} \sum_{i=1}^{p} E_i, \qquad z = z_{1-\alpha/2}. \tag{16}$$

Theorem 3 (Fixed-radius affine intervals)**.** *Suppose the honesty class contains 0 and all $\pm M E_i$. If $0 < \alpha < 1/2$ and*

$$C(Z) = [\langle W, Z\rangle + b - h, \langle W, Z\rangle + b + h]$$

is honest, with deterministic W, b, h, then

$$h \ge \frac{(M/\sqrt{p})z\tau}{M/\sqrt{p} + z\tau}. \tag{17}$$

On the rank-one tensor Frobenius ball, the best radius in this affine class is therefore of order $\min(M/\sqrt{p}, \tau)$, up to constants depending on α.

The theorem allows W to use the entire tensor. It does not cover affine centers with data-dependent radii. The matching upper order follows by choosing between I_G and I_0 for rank cap one.

For a nonempty proper subset S of modes, write $A_{[S]}$ for the corresponding bipartition unfolding and define

$$\Theta_S = \{A : \operatorname{rank}(A_{[S]}) \le 1,\ \|A\|_F \le M\}. \tag{18}$$

Theorem 4 (Cost of a single unfolding honesty class)**.** *If $0 < \alpha < 1/4$, every interval honest on Θ_S satisfies*

$$\mathbb{E}_0|C| \ge c_\alpha \min\left\{\frac{M}{\sqrt{p}}, \frac{\tau}{4}\right\}, \qquad c_\alpha = 1 - 2\alpha - \frac{1}{2}\sqrt{\sqrt{8/7} - 1} > 0. \tag{19}$$

The lower order $\min(M/\sqrt{p}, \tau)$ is attained up to α-dependent constants by the direct/structural choice on Θ_S.

The alternatives in the proof are rank one in the chosen matrix unfolding. They generally violate the other tensor rank constraints. This is necessary for the comparison: the theorem measures the cost of requiring coverage on the larger class. For Tucker caps, a mode unfolding is a permissible relaxation; for TT caps, a contiguous cut is permissible. With all tensor rank caps one, every bipartition in (18) contains the original tensor class. Increasing an unfolding cap further preserves the lower bound.

Theorem 5 (Strict length separations). *Let $0 < \alpha < 1/4$, $K \geq 3$ and the rank caps be fixed. Under (16) and $M = \tau\sqrt{p}$, C_* is honest on the Tucker or TT class and*

$$|C_*| \leq 2\tau J^{1-1/K} \left\{ \frac{8\log(2p/\alpha)}{p} \right\}^{(K-2)/(2K)} = o(\tau). \tag{20}$$

In contrast, every fixed-radius affine interval honest on the same class has length at least $2z\tau/(1+z)$. Every interval honest on a class retaining only one constrained unfolding has expected length at zero at least $c_\alpha \tau/4$.

For growing caps, the same conclusion follows if $J^{1-1/K}(\log p/p)^{(K-2)/(2K)} \to 0$. This condition may be restrictive. At $K = 2$ the exponent in (20) is zero, so the argument gives no polynomial separation. Theorem 5 neither asserts that the displayed exponent is optimal nor compares favorably at every strong signal with an efficient oracle projection interval. The following benchmark concerns the diagonal observation experiment; the subsequent full-data theorem gives the final rate.

3.3 An initial full-data lower bound

Theorem 6 (Within-class lower bound). *Under (1) and (16), for $0 < \alpha < 1/4$, every interval honest on a class containing the rank-one tensor Frobenius ball of radius M satisfies*

$$\mathbb{E}_0|C| \geq (3/4 - 2\alpha)\frac{\min\{M, \tau\sqrt{\log(1+p/4)}\}}{\sqrt{p}}. \tag{21}$$

This is a sparse-coordinate mixture bound. At $M = \tau\sqrt{p}$, for fixed caps and fixed $K \geq 3$, the results imply

$$c\tau\sqrt{\frac{\log p}{p}} \leq \mathcal{L}_0(\Theta) \leq \mathcal{L}_{\max}(\Theta) \leq C\tau\left(\frac{\log p}{p}\right)^{(K-2)/(2K)} \tag{22}$$

for sufficiently large p, with constants depending on K, the caps and α. The ratio of these initial scales grows as $(p/\log p)^{1/K}$. Section 3.6 closes the gaps: the two criteria in (7) have the same rate, with an additional logarithmic factor in the cubic case.

3.4 Optimality when only the diagonal is observed

Let Θ_1 denote the Frobenius ball of tensors with rank at most one, including zero. Restrict the interval to be measurable in $d = (Z_{i,\dots,i})_{i=1}^p$ and an independent random seed. Write $\mathcal{C}_{\alpha,d}(\Theta_1)$ for such intervals that are honest on Θ_1, and define

$$\mathcal{L}_{0,d}(\Theta_1) = \inf_{C \in \mathcal{C}_{\alpha,d}(\Theta_1)} \mathbb{E}_0|C|, \qquad \mathcal{L}_{\max,d}(\Theta_1) = \inf_{C \in \mathcal{C}_{\alpha,d}(\Theta_1)} \sup_{A \in \Theta_1} \mathbb{E}_A|C|.$$

The next result specializes classical sparse-sequence arguments [Cai and Low, 2004, Collier et al., 2017] to honest intervals. It is a benchmark for the proposed use of diagonal information, not a new Gaussian-sequence minimax principle.

Proposition 1 (Classical diagonal-information benchmark). *Assume (1), (16) and $0 < \alpha < 1/4$. For each integer $1 \leq s \leq p$, put*

$$b_s = \min\left\{ Ms^{-K/2},\ \tau\sqrt{\log\left(1 + \frac{p}{4s^2}\right)} \right\}, \qquad c_{\alpha,d} = 1 - 2\alpha - \frac{1}{2}\sqrt{e^{1/4} - 1} > 0.$$

Every $C \in \mathcal{C}_{\alpha,d}(\Theta_1)$ satisfies

$$\mathbb{E}_0|C| \geq c_{\alpha,d}\frac{sb_s}{\sqrt{p}}. \tag{23}$$

For fixed $K \geq 3$ and $M = \tau\sqrt{p}$,

$$\mathcal{L}_{0,d}(\Theta_1) \asymp \mathcal{L}_{\max,d}(\Theta_1) \asymp \tau \left(\frac{\log p}{p}\right)^{(K-2)/(2K)}. \tag{24}$$

The comparison constants depend on K and α.

The rate in (24) follows because the effective support size $s \asymp (p/\log p)^{1/K}$ makes (23) match the threshold interval's upper order. The same asymptotic conclusion holds for any fixed Tucker or TT caps: the class contains Θ_1 and Theorem 2 changes the upper bound only by a fixed rank factor. Thus further improvement in the power of p cannot come from recalibrating a procedure restricted to d.

For comparison with the full-data result, consider the preliminary bound in (22). For the common-target tensors $A_S = b_s(\sum_{i\in S} e_i)^{\otimes K}$, two support sets with overlap H have diagonal inner product $b_s^2 H$ but full-tensor inner product $b_s^2 H^K$. The likelihood second moment in the full experiment is consequently different. A valid diagonal lower bound cannot be transferred by ignoring the extra observations. Theorem 7 establishes that they do permit shorter expected length at zero.

Table 2 separates the procedure and parameter classes in the comparisons.

Comparison	Information and honesty class	Length statement when $M = \tau\sqrt{p}$
Direct Gaussian interval	All means; prescribed L	$2z\tau$, with exact coverage.
Selected interval	Joint cap class; uses d	At most $2\tau J^{1-1/K}\{8\log(2p/\alpha)/p\}^{\beta}$; $\beta = (K-2)/(2K)$.
Any fixed-radius affine interval	Full tensor; same joint class	Length at least $2z\tau/(1+z)$.
Any interval on one unfolding class	Full tensor; enlarged honesty class	Expected length at zero at least $c_\alpha\tau/4$.
Any diagonal-based interval	Only d; rank-one or fixed-cap class	Optimal expected length at zero and in worst case has order $\tau(\log p/p)^{\beta}$.
Any full-data interval	Full tensor; joint class	Optimal expected length both at zero and in the worst case is $r_K(p)$ in (34).

Table 2: Precisely delimited comparisons for fixed $K \geq 3$ and fixed rank caps. Lower bounds for relaxed honesty classes or restricted observations are not lower bounds for unrestricted full-data procedures on the original class.

3.5 Using the full tensor at the zero mean

The diagonal rate is a constraint on procedures using the diagonal. A finite scan over product directions of coordinate subtensors can give shorter expected length at zero while preserving the original uniform length bound. The scan is an existence construction with exponential complexity; no efficient implementation is asserted.

For a tensor B, define its injective norm

$$\rho(B) = \sup_{\|u_k\|_2=1} |\langle B, u_1 \otimes \cdots \otimes u_K\rangle|.$$

For every nonempty subset $S \subseteq \{1,\ldots,p\}$ of size s, choose fixed $1/(2K)$-nets of the unit sphere in $\mathbb{R}^s$, one per mode, each with at most $(1+4K)^s$ points. Let $N_S(Z)$ be the largest absolute contraction of $Z_{S,\ldots,S}$ over their product. Define

$$\lambda_s^{\text{scan}} = \tau\left[2\{s\log(ep/s) + Ks\log(1+4K) + \log(2p/\gamma)\}\right]^{1/2}, \tag{25}$$

$$R_\gamma = \frac{4J^{K/2}}{\sqrt{p}} \sum_{s=1}^{p} \lambda_s^{\text{scan}} s^{-K/2}. \tag{26}$$

Accept the zero test if $N_S(Z) \le \lambda^{\mathrm{scan}}_{|S|}$ for every nonempty S. On acceptance, let $C_{\mathrm{scan}} = [-h_{\mathrm{scan}}, h_{\mathrm{scan}}]$, where $h_{\mathrm{scan}} = \min(h_0, R_\gamma)$; on rejection let $C_{\mathrm{scan}} = [-h_0, h_0]$. Here $h_0 = M\sqrt{s_{\mathrm{cap}}/p}$ and s_{cap} is the smallest cap, truncated at p. For $0 < \alpha_0, \gamma < 1$ with $\alpha_0 + \gamma \le \alpha$, intersect this interval with $C_*(\alpha_0)$ and call the result $C_\dagger$; replace an empty intersection by $\{0\}$. Write $\ell_* = |C_*(\alpha_0)|$.

Theorem 7 (A full-data interval with short expected length at zero)**.** *Under* (1), (16) *and either cap class,* C_{scan} *is honest at level* $1-\gamma$*, and* $C_\dagger$ *is honest at level* $1-\alpha$*. Moreover,*

$$\sup_{A\in\Theta} \mathbb{E}_A|C_\dagger| \le \ell_*, \qquad \mathbb{E}_0|C_\dagger| \le \min\{\ell_*,\, 2R_\gamma + \ell_*\gamma\}. \tag{27}$$

For fixed $0 < \alpha < 1/4$*, fixed* $K \ge 3$*, fixed positive caps and* $M = \tau\sqrt{p}$*, choose* $\alpha_0 = \alpha/2$ *and* $\gamma = \alpha/(2p)$*. Then*

$$\mathbb{E}_0|C_\dagger| \le C_{K,J,\alpha}\frac{\tau\sqrt{\log p}}{\sqrt{p}}\sum_{s=1}^{p} s^{-(K-1)/2} \tag{28}$$

for sufficiently large p*. Consequently,*

$$\mathcal{L}_0(\Theta) \asymp \tau\sqrt{\frac{\log p}{p}}, \qquad K \ge 4. \tag{29}$$

The matching lower bound in (29) is Theorem 6, applied to the same full-data experiment and honesty class. The sum in (28) is bounded for $K \ge 4$ and harmonic for $K = 3$. The number of scanned product directions is bounded by $(1 + (1+4K)^K)^p - 1$. Thus the result establishes attainable statistical precision, without a polynomial-time guarantee. The worst-case bound in (27) is the initial diagonal upper order. The residual projection in Theorem 10 strengthens it to the sharp full-data rate.

Theorem 8 (A harmonic obstruction in order three)**.** *Under* (1) *and* (16)*, let* $K = 3$*,* $M = \tau\sqrt{p}$ *and* $0 < \alpha < 1/4$*. Every interval honest on a class containing the rank-one tensor Frobenius ball satisfies*

$$\mathbb{E}_0|C| \ge (3/4 - 2\alpha)\frac{\tau H_p}{200\sqrt{p}}, \qquad H_p = \sum_{i=1}^{p} i^{-1}. \tag{30}$$

Theorem 8 uses a full-data mixture of rank-one tensors with sign-coupled factors and a common harmonic diagonal. It refutes the earlier conjecture that $\mathcal{L}_0$ always has order $\tau\sqrt{\log p/p}$ at rank one. Combining (30) and (28) gives the intermediate cubic bracket

$$c^{\mathrm{cub}}_\alpha \frac{\tau \log p}{\sqrt{p}} \le \mathcal{L}_0(\Theta) \le C_{J,\alpha}\frac{\tau(\log p)^{3/2}}{\sqrt{p}}. \tag{31}$$

Here $c^{\mathrm{cub}}_\alpha = (3/4 - 2\alpha)/200 > 0$. The intermediate bracket (31) has a factor $\sqrt{\log p}$ gap. Theorem 9 closes it from below. The constant $1/200$ is conservative and is not an optimal finite-sample constant.

Further verified refinements. The accompanying theory review proves a smaller rank coefficient in the diagonal bound, an exact one-dimensional characterization of the optimal fixed affine radius, and a noise-envelope version with an additive approximation term. It also records finite-variance and unknown-Gaussian-variance extensions requiring retained replicates. These supporting results are collected in Appendix C; the simulations retain the original Gaussian threshold procedure.

3.6 Sharp full-data rates, including worst-case length

A sparse harmonic profile strengthens the cubic lower bound. Inverting the residual scan then turns the zero-point construction into an interval with uniformly controlled length.

Theorem 9 (Sparse harmonic lower bound)**.** *Under (1) and (16), let $K = 3$ and $0 < \alpha < 1/4$. For every integer $1 \le m \le p$, put*

$$b_m = \min\left\{\frac{M}{\sqrt{27m}}, \frac{\tau}{\sqrt{27}}\sqrt{\log(1 + p/(4m^2))}\right\}, \qquad c_{\alpha,d} = 1 - 2\alpha - \tfrac{1}{2}\sqrt{e^{1/4} - 1}.$$

Every interval honest on a class containing the rank-one Frobenius ball satisfies

$$\mathbb{E}_0|C| \ge c_{\alpha,d}\frac{b_m H_m}{\sqrt{p}}. \tag{32}$$

If $M = \tau\sqrt{p}$, then for $p \ge 256$ and $m = \lfloor p^{1/4} \rfloor$,

$$\mathbb{E}_0|C| \ge \frac{c_{\alpha,d}}{16\sqrt{27}}\frac{\tau(\log p)^{3/2}}{\sqrt{p}}.$$

The profile weights, rather than additional rank components, produce the harmonic sum. The alternatives are nonnegative rank-one tensors. Their full likelihood overlap is bounded linearly in the support intersection, permitting the classical hypergeometric calculation.

Write $\lambda_s(\gamma) = \lambda_s^{\mathrm{scan}}$ for the threshold in (25) at error budget γ. Use the product nets above with thresholds $\lambda_s(\alpha)$. Define

$$\mathcal{F}(Z) = \{B \in \Theta(M, \bar{r}) : N_S(Z - B) \le \lambda_{|S|}(\alpha) \text{ for every nonempty } S\}.$$

Let C_F be the interval between the minimum and maximum of ψ on $\mathcal{F}(Z)$, and set $C_F = \{0\}$ when this set is empty. Write $J_2 = J(2\bar{r})$, the expansion constant with doubled caps, and put

$$W_\alpha = \frac{4J_2^{K/2}}{\sqrt{p}}\sum_{s=1}^{p}\lambda_s(\alpha)s^{-K/2}.$$

Theorem 10 (Uniform diameter and full-data minimax length)**.** *Under (1), (16), $0 < \alpha < 1$ and either positive cap class, C_F is measurable and honest at level $1 - \alpha$, and for every observation*

$$|C_F| \le \min(2h_0, W_\alpha). \tag{33}$$

For fixed $0 < \alpha < 1/4$, fixed $K \ge 3$, fixed positive caps and $M = \tau\sqrt{p}$,

$$\mathcal{L}_0(\Theta) \asymp \mathcal{L}_{\max}(\Theta) \asymp r_K(p), \qquad r_K(p) = \begin{cases} \tau(\log p)^{3/2}/\sqrt{p}, & K = 3, \\ \tau\sqrt{\log p/p}, & K \ge 4. \end{cases} \tag{34}$$

Constants may depend on K, the caps and α.

Differences of feasible tensors obey doubled unfolding caps; the proof does not assume that a rank class is closed under subtraction. Computing the confidence-set extrema is not claimed to be efficient.

Proposition 2 (Squared-error consequence)**.** *In the regime of (34),*

$$\inf_{\widehat{\psi}}\sup_{A\in\Theta}\mathbb{E}_A(\widehat{\psi} - \psi(A))^2 \asymp r_K(p)^2.$$

The midpoint of C_F with error level $\alpha_p = p^{-2}$ attains the upper order for $p \ge 2$.

3.7 Computation: a certified hierarchy and a restricted lower bound

The exact residual projection supplies the statistical benchmark. A truncated zero test is computable with an explicit search budget. It does not compute the extrema defining C_F.

Proposition 3 (Truncated scans)**.** *For $1 \le t \le p$, test zero using only supports of sizes at most t, retaining thresholds calibrated simultaneously over all sizes. Write $\lambda_s = \lambda_s(\gamma)$ and put*

$$q = 2/K, \quad B_q = J^{1-1/K} M^q, \quad a_t = 4J^{K/2}\lambda_t t^{-K/2},$$

$$R_t = \frac{4J^{K/2}}{\sqrt{p}} \sum_{s=1}^{t} \lambda_s s^{-K/2} + \mathbf{1}_{\{t<p\}} \frac{B_q}{\sqrt{p}} a_t^{1-q}.$$

On acceptance use radius $\min(h_0, R_t)$ centered at zero; on rejection use the structural interval. Intersect with $C_*(\alpha_0)$, where $0 < \alpha_0, \gamma < 1$, $\alpha_0 + \gamma \le \alpha$, and γ is the scan error budget, using the singleton convention. The resulting C_t is honest, has length at most $\ell_* = |C_*(\alpha_0)|$, and

$$\mathbb{E}_0|C_t| \le \min(\ell_*, 2R_t + \ell_*\gamma).$$

For fixed $K \ge 3$, fixed caps and fixed $0 < \alpha < 1$, with $M = \tau\sqrt{p}$, $\alpha_0 = \alpha/2$ and $\gamma = \alpha/(2p)$,

$$R_t \lesssim \frac{\tau\sqrt{\log p}}{\sqrt{p}} \sum_{s=1}^{t} s^{-(K-1)/2} + \mathbf{1}_{\{t<p\}} \tau (\log p/p)^{\beta} t^{-(K-1)\beta}, \qquad \beta = (K-2)/(2K). \tag{35}$$

Explicit normalized grid nets have at most $(3+4K\sqrt{s})^s$ directions per mode. Using their cardinalities in the Gaussian thresholds preserves (35). The arithmetic cost through level t is bounded by

$$O\left\{\sum_{s=1}^{t} \binom{p}{s} (3+4K\sqrt{s})^{Ks} s^K\right\}.$$

For fixed t, K this is polynomial in p. Fixed t does not close the power gap in the tail term of (35). Completed levels share one noise event, so successive intersections remain honest even if the decision to compute another complete level uses the data.

A complementary obstruction holds for polynomial estimators under squared-error loss. This is the restricted low-degree framework used in sparse tensor PCA [Choo and d'Orsi, 2021, Fu and Sohn, 2026]; it does not characterize all polynomial-time computation.

Theorem 11 (A low-degree estimation separation)**.** *Let $K \ge 3$, $1 \le s \le \sqrt{p}$, and mix uniformly over*

$$x_S = s^{-1/2}\mathbf{1}_S, \qquad A_S = \tau\lambda x_S^{\otimes K}, \qquad |S| = s.$$

Assume $0 < \tau\lambda \le M$, and write

$$\delta = \frac{\tau\lambda s^{1-K/2}}{\sqrt{p}}, \quad \mu = s^2/p, \quad B_{\mathsf{d}} = \frac{\lambda^2}{s^K}(K\mathsf{d})^K,$$

where $\mathsf{d} \ge 1$ is the polynomial degree. If $B_{\mathsf{d}} < 1$, the centered likelihood ratio projected onto polynomials of degree at most d has squared null norm at most

$$\eta_{\mathsf{d}}^2 = \mu e^{\mu} B_{\mathsf{d}}/(1 - B_{\mathsf{d}}).$$

Every scalar polynomial estimator $T(Z)$ of degree at most d satisfies

$$\sup_{A\in\Theta} \mathbb{E}_A(T-\psi(A))^2 \geq \frac{\delta^2}{(2+\eta_{\mathsf{d}})^2} \tag{36}$$

on any parameter class containing zero and these alternatives. In particular, let $M=\tau\sqrt{p}$, $\lambda=\sqrt{p}$, $s=\lfloor p^a\rfloor$, and $\mathsf{d}=\lfloor p^b\rfloor$, where

$$\frac{1}{K}<a<\min\left(\frac{1}{2},\frac{1}{K-2}\right), \qquad 0<b<a-\frac{1}{K}.$$

Then $\eta_{\mathsf{d}}\to 0$ and the lower order in (36) is polynomially larger than the unrestricted rate in Proposition 2.

This theorem includes $K=3$ and uses nonnegative sparse factors. It does not imply the same limitation for arbitrary interval endpoints or arbitrary thresholded algorithms.

Proposition 4 (What an efficient short interval would imply)**.** *Let Q be supported on parameters with common target $\delta>0$, and let the null target be zero. An interval honest at level $1-\alpha$ at the null and these alternatives defines $\phi(Z)=\mathbf{1}\{\delta\in C(Z)\}$ with*

$$Q(\phi=0)\leq\alpha, \qquad P_0(\phi=1)\leq\alpha+\mathbb{E}_0|C|/\delta.$$

A computational hypothesis excluding polynomial-time tests with total error below $c>2\alpha$ therefore excludes polynomial-time honest intervals with $\mathbb{E}_0|C|<(c-2\alpha)\delta$ for that prior.

The last implication is conditional on a testing-hardness hypothesis. Theorem 11 alone does not establish that hypothesis. Efficient minimax confidence intervals remain open, with a proved restricted estimation obstruction and an explicit computable hierarchy.

4 Norm budgets and loading geometry

The restriction $M=\tau\sqrt{p}$ can be removed for the equal diagonal contrast. The key is to impose the Frobenius budget and the residual-scan constraints on each ordered diagonal entry, before summing. Throughout this section the tensor order and positive rank caps are fixed; constants are not claimed uniform in growing caps.

Put $P_2=2J_2^{(K-1)/2}$ and $Q_2=4J_2^{K/2}$, where $J_2=J(2\bar{r})$, and define

$$U_{K,p}(M;\alpha)=\frac{1}{\sqrt{p}}\sum_{s=1}^{p}\min\{P_2M,Q_2\lambda_s(\alpha)\}\,s^{-K/2}. \tag{37}$$

Here $\lambda_s(\alpha)$ is the finite-net threshold in (25), with error budget α.

Theorem 12 (Diameter using both norm and scan constraints)**.** *For every $p\geq 1$, $K\geq 3$, $M,\tau>0$ and $0<\alpha<1$, the residual-projection interval C_F is honest and satisfies, for every observation,*

$$|C_F|\leq\min\{2h_0,U_{K,p}(M;\alpha)\}.$$

The bound retains the supplied norm budget inside the ordered-coordinate sum; taking the minimum only after summation can lose the intermediate cubic regime.

Let $\ell_p=\log(ep)$, $x=M/(\tau\sqrt{\ell_p})$, and define

$$g_p(x)=\min\{x,1+\log_+ x,\ell_p\}, \qquad \log_+ x=\max(0,\log x), \tag{38}$$

$$\mathfrak{r}_K(p,M,\tau)=\frac{\tau\sqrt{\ell_p}}{\sqrt{p}}\begin{cases} g_p(x), & K=3,\\ \min(x,1), & K\geq 4.\end{cases} \tag{39}$$

These definitions also cover $p=1$, when the rate is $\min(M,\tau)$.

Theorem 13 (Full-data length for every positive norm budget)**.** *Fix $K \geq 3$, positive Tucker or TT caps, and $0 < \alpha < 1/4$. For the equal diagonal contrast there are constants $0 < c < C < \infty$, depending only on these fixed quantities, such that for every $p \geq 1$ and every $M, \tau > 0$,*

$$c\,\mathfrak{r}_K(p, M, \tau) \leq \mathcal{L}_0(\Theta) \leq \mathcal{L}_{\max}(\Theta) \leq C\,\mathfrak{r}_K(p, M, \tau).$$

The lower bounds already hold when the honesty class is restricted to rank-one tensors. The interval in Theorem 12 attains the upper order.

For $K \geq 4$ the rate is $\min\{M, \tau\sqrt{\log(ep)}\}/\sqrt{p}$. For $K = 3$, the norm-limited regime $x \leq 1$ has rate $M/\sqrt{p}$; above it the multiplier grows as $1 + \log x$ until capped at ℓ_p. These are rates up to fixed multiplicative constants, not exact transition constants. Substituting $M = \tau\sqrt{p}$ recovers Theorem 10.

For squared-error estimation, a fixed-confidence midpoint argument is insufficient when M/τ is arbitrarily large: its off-coverage term can grow with M^2. Instead use an approximate minimum-residual estimator. Fix $\beta = 1/8$, and put

$$T(R) = \max\bigl(0, \max_{S \neq \varnothing}\{N_S(R) - \lambda_{|S|}(\beta)\}\bigr).$$

Choose a fixed countable dense sequence (B_j) in the compact cap class. Define

$$d(Z) = \inf_j T(Z - B_j), \quad j(Z) = \min\{j : T(Z - B_j) \leq d(Z) + \tau\}, \quad \widehat{\psi} = \psi(B_{j(Z)}). \tag{40}$$

The tolerance τ ensures the displayed index exists. This defines a statistical estimator, not a claimed efficient optimization algorithm.

Theorem 14 (Squared risk uniformly over norm budgets)**.** *Under the fixed-order, fixed-cap conditions of Theorem 13,*

$$\inf_{\widetilde{\psi}} \sup_{A \in \Theta} \mathbb{E}_A(\widetilde{\psi} - \psi(A))^2 \asymp \mathfrak{r}_K(p, M, \tau)^2$$

uniformly over $p \geq 1$ and $M, \tau > 0$. The measurable estimator in (40) attains the upper bound. It uses neither a failure probability depending on M nor a minimum nonzero singular value.

4.1 Nonuniform orthogonal loadings

Return to the prescribed orthogonal product loading in (4), with m nonzero weights and possibly unequal ambient mode dimensions. Let w_s^* denote decreasing absolute weights. Use orthogonal mode transformations and restrict to the m^K coordinates spanned by the prescribed loading vectors. The resulting Gaussian tensor retains the rank and norm bounds. Write $\lambda_s^{(m)}$ for the scan threshold with $p = m$, and define

$$U_w = \sum_{s=1}^{m} w_s^* \min\{P_2 M, Q_2 \lambda_s^{(m)}(\alpha)\} s^{-K/2}, \tag{41}$$

$$L_w = \max_{1 \leq d \leq m} w_d^* \sqrt{d}\,\mathfrak{r}_K(d, M, \tau). \tag{42}$$

The rank constants use the same supplied caps. The structural radius is

$$h_0 = M \left(\sum_{s=1}^{s_{\mathrm{cap}}} (w_s^*)^2 \right)^{1/2},$$

with s_{cap} truncated at m.

Theorem 15 (Loading-dependent full-data bounds)**.** *For fixed $K \geq 3$, fixed positive caps and fixed $0 < \alpha < 1/4$, an honest interval obtained by residual projection in the rotated block satisfies*

$$|C_w(Z)| \leq \min(2h_0, U_w)$$

for every observation. Conversely every interval honest on the full ambient cap class satisfies

$$\mathbb{E}_0|C(Z)| \geq c_{K,\alpha} L_w.$$

The lower bound allows use of every ambient tensor entry. These bounds are not asserted to match for arbitrary nonuniform weights.

Proposition 5 (Sharp rates for a flat orthogonal loading)**.** *If $w_i = \epsilon_i/\sqrt{m}$ with prescribed $\epsilon_i \in \{-1, 1\}$, then both minimax expected-length criteria have order $\mathfrak{r}_K(m, M, \tau)$, and minimax squared error has its square, independently of the larger ambient mode dimensions. The statement includes dense orthogonal rotations of the loading. At $m = 1$ it gives the bounded-normal-mean scale* $\min(M, \tau)$.

Scope of the general-loading bounds. The displayed envelopes need not match for arbitrary unequal weights. Results L1 and W2 below establish additional small-budget and fixed-decay regimes. Results A1, G2 and P1 separately describe fixed-cap rate adaptation, a growing-outer-class obstruction and strong saturated pointwise inference. General loading profiles and complete pointwise adaptation remain open.

5 Robust noise, computation, and rank adaptation

This section separates extensions of validity from sharp minimax claims. It also separates adaptation on a spectrally separated class from honesty on the original class containing arbitrarily weak components. Lower-bound comparisons use a fixed error level below 1/4. All statements have proofs in Appendix F.

5.1 Beyond Gaussian observations

For a centered error tensor E, the following joint directional condition is sufficient:

$$\mathbb{E}\exp(t\langle U, E\rangle) \leq \exp(t^2 v^2 \|U\|_F^2/2) \quad (U \text{ deterministic},\ t \in \mathbb{R}). \tag{43}$$

It allows dependence and nonidentical coordinates. Independent centered sub-Gaussian coordinates with common proxy v satisfy it, as does Gaussian covariance bounded above by $v^2 I$. Marginal coordinate bounds alone do not imply it.

Theorem 16 (Full-data robustness under directional tails)**.** *In $Z = A + E$, replace τ by a known $v > 0$ in the scan thresholds and residual-fit tolerance. Under (43), the norm-sensitive diameter, uniform squared-risk upper bound, and orthogonal-loading upper bounds remain valid. For fixed order/caps, their equal-diagonal orders are $\mathfrak{r}_K(p, M, v)$ and its square. If the noise law is an unknown member of the family satisfying (43), these rates are minimax when coverage and the supremum risk/length range over that family. This lower-bound statement need not hold for each fixed non-Gaussian law separately.*

For a finite-variance extension retain independent identically distributed full-tensor replicates $W_1, \ldots, W_n$, with mean A and $\mathrm{Cov}(\mathrm{vec}\, W_j) \preceq \sigma^2 I$ for known $\sigma > 0$. Put $\tau = \sigma/\sqrt{n}$, and let

$$H_s = s\log(ep/s) + Ks\log(1 + 4K).$$

For a desired error $0 < \beta < 1/4$, choose b_s as the smallest odd integer at least

$$\max\{9, [H_s + \log(p/\beta)]/\log 2\}, \qquad n \geq 2\max_s b_s. \tag{44}$$

For each unit product-net direction U supported on S^K, $|S| = s$, let d_U be the median of its b_s disjoint block averages, using block size $k_s = \lfloor n/b_s \rfloor$. The block partitions may differ across s but are deterministic. Set $\lambda_s^{\mathrm{mom}} = 4\sigma/\sqrt{k_s}$.

Theorem 17 (Full scans with only finite-variance replicates)**.** *Use the feasible constraints $|d_U - \langle U, B\rangle| \le \lambda_s^{\mathrm{mom}}$ in place of the Gaussian residual constraints. Their projection onto the equal-diagonal target is honest at level $1-\beta$, with the norm-sensitive diameter using λ_s^{mom}. For fixed K, caps and confidence, both minimax expected-length criteria, with a supremum over all the stated replicate laws, have order $\mathfrak{r}_K(p, M, \sigma/\sqrt{n})$ whenever (44) holds.*

For squared error use $\beta = 1/8$ and approximately minimize

$$F(B) = \max\bigl(0, \max_U\{|d_U - \langle U, B\rangle| - \lambda_{s(U)}^{\mathrm{mom}}\}\bigr)$$

over a fixed countable dense cap-class sequence, within tolerance τ. The resulting scalar estimator has minimax squared risk of order $\mathfrak{r}_K^2$ over the same family. This is an existence construction. The sufficient replicate count is $n \ge C_{K,\beta} p$, and the full directional search remains exponential.

The moments d_U need not be contractions of a single tensor estimate. Their separate linear constraints suffice. No Gaussian likelihood is used for the robust upper bounds; the lower bounds use the Gaussian subfamily of replicate distributions. This is a finite-direction use of the classical median-of-means principle [Devroye et al., 2016].

Proposition 6 (Two limits of a noise relaxation)**.** *(i) In one dimension, known noise taking values $\pm v$ with equal probabilities permits exact recovery of every mean $|A| \le M < v$, although it has variance v^2 and satisfies (43). Thus a Gaussian lower bound cannot be transferred to every noise law from a variance bound alone. (ii) For $E = v\xi\mathbf{1}$ with one Rademacher sign ξ, every coordinate has proxy v, but a flat unit product contraction on a p^K block has variance $v^2 p^K$. Coordinatewise calibration cannot certify a full scan or a dense rotated loading.*

5.2 Approximate ranks and growing caps

Theorem 18 (Sharp length under a Frobenius approximation allowance)**.** *In the original Gaussian experiment let $\Theta^{\varepsilon_A} = \Theta(M, \bar{r}) + \varepsilon_A \mathbb{B}_F$, $\varepsilon_A \ge 0$. For the equal diagonal, fixed $K \ge 3$, fixed positive caps and fixed $0 < \alpha < 1/4$,*

$$\mathcal{L}_0(\Theta^{\varepsilon_A}) \asymp \mathcal{L}_{\max}(\Theta^{\varepsilon_A}) \asymp \min\{\tau, \mathfrak{r}_K(p, M, \tau) + \varepsilon_A\}.$$

The upper interval inflates every scan threshold by ε_A and both projected endpoints by ε_A, with a deterministically selected direct Gaussian fallback. Thus approximate low rank is allowed, but the original rate requires $\varepsilon_A = O(\mathfrak{r}_K)$.

Proposition 7 (A growing-rank lower bound and the unrestricted endpoint)**.** *Let r_* be the minimum supplied cap, truncated at p. Under equal loading every honest interval obeys*

$$\mathbb{E}_0|C| \ge \frac{c_\alpha}{\sqrt{p}} \max_{1 \le s \le r_*} \min\{M\sqrt{s}, \tau s\sqrt{\log(1 + p/(4s^2))}\}.$$

The finite upper envelopes of the preceding sections remain valid for dimension-dependent supplied caps with their displayed cap constants; these constants are not uniformly bounded. If all rank restrictions are removed, then for any prescribed unit loading both length criteria have order $\min(M, \tau)$, attained in polynomial time. The vanishing tensor rate therefore does not persist under unrestricted ranks.

5.3 Arbitrary loading tensors and efficient regimes

For a nonzero, otherwise arbitrary deterministic tensor L, a useful sufficient description is a certified decomposition $L = L_0 + R$, where L_0 has a tractable orthogonal loading representation and the remainder R is known exactly. Normalization is not required in this subsection.

Proposition 8 (A confidence interval for any certified loading decomposition)**.** *Suppose $C_0 = [a_0, b_0]$ is honest for $\langle L_0, A\rangle$ at error α_0, and $\mathbb{P}(|\langle R, E\rangle| > q_R) \le \alpha_1$ uniformly. Then*

$$C_L = [a_0 + \langle R, Z\rangle - q_R,\ b_0 + \langle R, Z\rangle + q_R]$$

is honest at error $\alpha_0 + \alpha_1$ and has length $|C_0| + 2q_R$. Under (43), one may take

$$q_R = v\|R\|_F\sqrt{2\log(2/\alpha_1)}.$$

For Gaussian noise use the normal quantile. No independence between the two terms is required. A decomposition or partition selected from loading and supplied constants alone needs no selection penalty. Selection using Z does require simultaneous calibration.

This also supplies additional unequal-weight upper bounds: partition the orthogonal terms into prescribed blocks, apply their separate weighted bounds with error budgets summing to α, and add the intervals. Include the unsplit construction among the choices. The minimum among their deterministic diameter bounds is achievable by choosing the construction before observing data.

Theorem 19 (Two efficiently attained minimax regimes)**.** *(i) For the equal diagonal, fixed order/caps and $M \le \tau\sqrt{\log(ep)}$, the structural interval is minimax up to fixed constants, with length of order $M/\sqrt{p}$. (ii) Let a parameter class contained in $\{A : \|A\|_F \le M\}$ contain every rank-one tensor of Frobenius norm at most M. If a supplied unit product tensor U certifies $|\langle L, U\rangle| \ge c\|L\|_F$ for a fixed $c > 0$, then under Gaussian observations both length criteria have order $\|L\|_F \min(M, \tau)$. Choose between the norm-only structural interval and the direct Gaussian interval. The algorithm uses a contraction and scalar comparisons, without rank recovery or tensor optimization. This covers arbitrary loadings with a certified constant product overlap, regardless of supplied rank caps.*

The general loading transfer is a valid upper bound, not a characterization of the optimum for every tensor L. The near-flat difficult contrasts do not satisfy a dimension-independent product-overlap condition.

Proposition 9 (Why full convexification loses the weak-signal gain)**.** *Any convex parameter set containing the rank-one Frobenius ball of radius M also contains $\{tL : |t| \le M/\sqrt{p}\}$ for the equal diagonal loading. Consequently, every interval honest on that convex set satisfies*

$$\mathbb{E}_0|C| \ge c_\alpha \min(M/\sqrt{p}, \tau).$$

At $M = \tau\sqrt{p}$ this is order τ, exceeding the original full-data rate by an unbounded factor as $p \to \infty$. The conclusion concerns procedures guaranteeing coverage on the whole convex outer class; it is not a computational lower bound for every algorithm using a convex subroutine.

5.4 Actual rank: an obstruction and a computable separated case

Theorem 20 (An adaptation obstruction at nonzero tensors)**.** *Suppose A_0 is supported on $\{1, \dots, q\}^K$, $q < p$, has norm $a < M$, and every supplied unfolding cap is at least its actual rank plus one. Put $m = p - q$ and $M_0 = (M^2 - a^2)^{1/2}$. For the equal diagonal every interval honest on the outer cap class satisfies*

$$\mathbb{E}_{A_0}|C| \ge c_{K,\alpha}\sqrt{m/p}\,\mathfrak{r}_K(m, M_0, \tau).$$

Thus arbitrarily weak extra components obstruct shortening at some nonzero lower-rank tensors, not only at zero. This is a lower bound, not a complete characterization of adaptation on all rank subclasses.

An explicit positive result uses two independent observations $Z^{(j)} = A + E^{(j)}$, with directional proxies v_j as in (43). Let $d_k = \prod_{\ell \neq k} p_\ell$, choose $0 < \delta < \alpha$, and set

$$\eta_k = 2v_1 \sqrt{2\{(p_k + d_k)\log 9 + \log(2K/\delta)\}}. \tag{45}$$

The allowed class has $\|A\|_F \leq M$ and, in each mode, either rank zero or smallest nonzero singular value strictly greater than $2\eta_k$. No ranks or rank caps are supplied to the algorithm. Separation is a model assumption; observed singular values alone cannot certify the absence of undetected weak components. Compute $\widehat{r}_k = \#\{j : \sigma_j(Z^{(1)}_{(k)}) > \eta_k\}$ and let Q_k project onto those left singular vectors. If any $\widehat{r}_k = 0$, return $\{0\}$. Otherwise put

$$d_k^* = \min\left\{1, \frac{2\eta_k}{\sigma_{\widehat{r}_k}(Z^{(1)}_{(k)}) - \eta_k}\right\}.$$

Here d_k^* is an error bound, distinct from the unfolding column dimension d_k. For the commuting mode projectors define the orthogonal projector

$$T = \bigotimes_k Q_k + \sum_k (I - Q_k) \otimes \bigotimes_{\ell \neq k} Q_\ell.$$

The tensor factors in this expression occupy their respective modes. This first-order product-space projector may be larger than the exact Tucker manifold tangent projector at ranks above one. Set

$$B_L = M\|(I - T)L\|_F \min\left\{1, \left(\sum_{k<\ell} (d_k^* d_\ell^*)^2\right)^{1/2}\right\}, \tag{46}$$

$$h_T = B_L + v_2\|TL\|_F \sqrt{2\log(2/(\alpha - \delta))}. \tag{47}$$

Theorem 21 (Finite-sample rank calibration with a computable bias correction)**.** *The interval centered at $\langle TL, Z^{(2)}\rangle$ with radius h_T is honest at level $1-\alpha$ on the separated class just specified, for any prescribed loading L. One may choose, using only the pilot and known quantities, the shortest-radius option among it, the direct second-observation interval with error $\alpha - \delta$, and the norm-only structural interval. It may also be intersected with the structural interval. For Gaussian second-observation noise the square-root logarithm can be replaced by $z_{1-(\alpha-\delta)/2}$.*

With probability at least $1 - \delta$, all selected mode ranks equal the actual ranks and (46) bounds the functional bias. The method consists of unfolding SVDs and tensor projections and is polynomial in the explicit input size. Its noise radius depends on $\|TL\|_F$, and its additional cost is the displayed bias term. General minimax optimality and adaptation without the spectral separation are not asserted.

For one Gaussian observation, independent artificial G' gives $Z^{(1)} = Z + \tau G'$ and $Z^{(2)} = Z - \tau G'$, with independent errors of scale $\sqrt{2}\tau$. For non-Gaussian errors, this trick need not give independence; genuinely independent replicate groups suffice. The rank statement concerns Tucker mode ranks, and applies to TT tensors meeting the mode-gap conditions, but does not identify their TT bond ranks. Subspace projection is established methodology [Xu et al., 2025]; the present contribution is an explicit conservative finite-sample certificate in this setting.

Assumption audit. Gaussian isotropy gives the exact diagonal sequence and the likelihood-ratio identities. The threshold interval alone extends whenever a valid simultaneous coordinate noise envelope is supplied, but the direct alternative and lower bounds require their own calibration. Orthonormal loading families produce Bessel bounds and equal noise variances; general loading results require the additional conditions stated later in this supplement. Canonical tensor expansions avoid a spectral gap but yield potentially large rank constants. Supplied caps and M identify the outer honesty class; replacing them with unvalidated estimates is not covered. None of these assumptions should be described as a routine approximation for dependent LLM representations.

A Complete proofs

The proofs apply in finite dimensions. Constants used in the main text are explicit. All tensors and Gaussian variables are real.

A.1 Canonical product expansions

Lemma 1 (Controlled squared component weights)**.** *For a tensor in either cap class there is a finite representation*

$$A = \sum_{j=1}^{J_A} x_{1j} \otimes \cdots \otimes x_{Kj}, \qquad a_j^{\mathrm{cp}} = \prod_{k=1}^{K} \|x_{kj}\|_2,$$

such that $J_A \le J$ *and* $\sum_j (a_j^{\mathrm{cp}})^2 = \|A\|_F^2$.

Roadmap. Choose orthonormal Tucker mode bases or left-orthogonal TT cores; expand product terms and sum their squared weights. The TT argument uses a prefix identity, not orthogonality of individual paths.

Proof. The zero tensor has a zero-term representation. For a nonzero Tucker tensor, choose orthonormal bases U_k for the column spaces of its mode unfoldings. Orthogonal projection onto each such space leaves A unchanged. Contracting these bases gives a core $\mathcal{G}$ with $A = \mathcal{G} \times_1 U_1 \cdots \times_K U_K$ and $\|\mathcal{G}\|_F = \|A\|_F$. The actual mode ranks satisfy $r_k \le \bar{r}_k$.

Fix a mode k. Expand all core indices other than k. Each tuple j_{-k} gives unit factors $U_\ell[:, j_\ell]$ in modes $\ell \ne k$ and factor $U_k \mathcal{G}[:, j_{-k}]$ in mode k. Its squared product weight is $\|\mathcal{G}[:, j_{-k}]\|_2^2$. These slice squares sum to $\|\mathcal{G}\|_F^2$. The number of terms is $\prod_{\ell \ne k} r_\ell$. Choosing k to minimize the corresponding cap product proves the Tucker assertion.

For a TT tensor, sequential matrix factorizations yield cores $G_k \in \mathbb{R}^{r_{k-1} \times p_k \times r_k}$, with $r_0 = r_K = 1$ and $r_k \le \bar{r}_k$. At each cut, factor the relevant matrix through a basis of its column space. Equivalently, left-orthogonalize the cores by QR and absorb the triangular factors into the next core. Thus the $(r_{k-1} p_k) \times r_k$ matrix associated with G_k has orthonormal columns for $k < K$, and $\|G_K\|_F = \|A\|_F$. These steps use no inverse lower bound on a singular value.

Expand the contraction over all internal paths $(a_1, \ldots, a_{K-1})$. A path contributes the product tensor with physical vectors $G_k[a_{k-1}, :, a_k]$. There are at most $\prod_{k<K} r_k \le J$ paths. To sum their squared weights, let

$$t_k(b) = \sum_{a_1, \ldots, a_{k-1}} \prod_{\ell=1}^{k} \|G_\ell[a_{\ell-1}, :, a_\ell]\|_2^2, \qquad a_k = b.$$

Column normalization gives $t_1(b) = 1$. If $t_{k-1}(a) = 1$ for all a, then

$$t_k(b) = \sum_a t_{k-1}(a) \|G_k[a, :, b]\|_2^2 = 1$$

for $k < K$. Hence the sum of all squared path weights is

$$\sum_a t_{K-1}(a) \|G_K[a, :, 1]\|_2^2 = \|G_K\|_F^2 = \|A\|_F^2.$$

The individual path terms need not be orthogonal; the prefix identity is what proves the required equality. □

A.2 Proof of Theorem 1

Roadmap. Apply subadditivity, modewise Hölder and Bessel, then control the component-weight sum.

Proof. Let $q = 2/K \le 1$ and use Lemma 1. Subadditivity of the qth power of absolute value gives

$$\sum_i |\langle A, V_i\rangle|^q \le \sum_j \sum_i \prod_{k=1}^{K} |\langle x_{kj}, v_{ki}\rangle|^{2/K}.$$

Hölder's inequality with K equal exponents, followed by Bessel's inequality for each orthonormal family, yields

$$\begin{aligned}\sum_i \prod_k |\langle x_{kj}, v_{ki}\rangle|^{2/K} &\le \prod_k \left(\sum_i |\langle x_{kj}, v_{ki}\rangle|^2\right)^{1/K} \\ &\le \prod_k \|x_{kj}\|_2^{2/K} = (a_j^{\mathrm{cp}})^q.\end{aligned}$$

Finally, the power-mean inequality (or Hölder with exponents $2/q$ and $2/(2-q)$) gives

$$\sum_j (a_j^{\mathrm{cp}})^q \le J_A^{1-q/2} \left\{\sum_j (a_j^{\mathrm{cp}})^2\right\}^{q/2} \le J^{1-q/2} M^q.$$

This is (14). For $J_A = 0$ both sides of the first inequality are zero. □

A.3 Exact rank-one diagonal image

Lemma 2 (Exact diagonal reduction)**.** *Under (4), the image of the rank-one tensor Frobenius ball of radius M under $A \mapsto (\langle A, V_i\rangle)_{i=1}^m$ is exactly*

$$\left\{a \in \mathbb{R}^m : \sum_i |a_i|^{2/K} \le M^{2/K}\right\}.$$

Roadmap. Use the diagonal bound for one inclusion and construct signed product factors for the converse.

Proof. One inclusion follows from Theorem 1 with $J = 1$. Conversely, for any vector in the displayed ball, put

$$x_1 = \sum_i \operatorname{sign}(a_i)|a_i|^{1/K} v_{1i}, \qquad x_k = \sum_i |a_i|^{1/K} v_{ki}, \quad k \ge 2,$$

and set $A = x_1 \otimes \cdots \otimes x_K$. Orthonormality gives $\langle A, V_i\rangle = a_i$ and

$$\|A\|_F = \prod_k \|x_k\|_2 = \left(\sum_i |a_i|^{2/K}\right)^{K/2} \le M.$$

The tensor has every Tucker and TT rank at most one. The zero vector is handled by the zero tensor. □

A.4 Threshold error and structural radius

Lemma 3 (Scalar threshold envelope)**.** *If $|e| \le \lambda$ and $s = \operatorname{soft}_\lambda(x + e)$, then s lies between zero and x and*

$$|s - x| \le \min(|x|, 2\lambda).$$

For $q \in (0, 1]$, $t, u \ge 0$, one also has $\min(t, u) \le t^q u^{1-q}$, with the continuous interpretation at zero.

Roadmap. Check positive and negative scalar inputs, then interpolate the two deterministic error bounds.

Proof. For $x \geq 0$, $x + e \geq -\lambda$, so thresholding cannot produce a negative value. Also $x + e - \lambda \leq x$, so $0 \leq s \leq x$. If $x > 2\lambda$, then $x + e > \lambda$ and $x - s = \lambda - e \leq 2\lambda$. Otherwise $x - s \leq x \leq 2\lambda$. The negative case follows by oddness. If $t \leq u$, then $t = t^q t^{1-q} \leq t^q u^{1-q}$; if $u \leq t$, then $u = u^q u^{1-q} \leq t^q u^{1-q}$. □

Lemma 4 (Matrix functional bound)**.** *If* $\operatorname{rank}(B) \leq r$ *and* H *has singular values* $s_1(H) \geq s_2(H) \geq \cdots$*, then*

$$|\langle H, B\rangle| \leq \|B\|_F \left(\sum_{i \leq r} s_i(H)^2\right)^{1/2}.$$

Roadmap. Project onto the matrix column space and maximize the spectral trace over projections of bounded rank.

Proof. Let P project onto the column space of B. Then $PB = B$ and Cauchy–Schwarz gives $|\langle H, B\rangle| = |\langle PH, B\rangle| \leq \|PH\|_F \|B\|_F$. Write $HH^\top = \sum_i \lambda_i u_i u_i^\top$, with decreasing nonnegative eigenvalues. The numbers $b_i = u_i^\top P u_i$ satisfy $0 \leq b_i \leq 1$ and $\sum_i b_i = \operatorname{tr}(P) \leq r$. Therefore $\|PH\|_F^2 = \sum_i \lambda_i b_i \leq \sum_{i \leq r} \lambda_i$: moving any weight from a smaller eigenvalue to an unfilled larger one cannot reduce this sum. Since $\lambda_i = s_i(H)^2$, the result follows. □

A.5 Proof of Theorem 2

Roadmap. Construct a uniform coordinate-noise event, bound the threshold error pathwise, verify the other intervals and use the deterministic selection rule.

Proof. The product atoms are orthonormal because

$$\langle V_i, V_j\rangle = \prod_k \langle v_{ki}, v_{kj}\rangle = \mathbf{1}\{i = j\}.$$

Gaussian isotropy then gives the independent normals in (5). For a standard normal g, Chernoff's inequality gives $\mathbb{P}(g > t) \leq \exp(-t^2/2)$ by minimizing $\exp(-ut + u^2/2)$ at $u = t$. Apply the same inequality to $-g$ and take a union bound to obtain

$$\mathbb{P}\{\max_i |\tau g_i| > \lambda\} \leq 2m \exp\{-\lambda^2/(2\tau^2)\} = \alpha.$$

On the complementary event, Lemma 3 and Theorem 1 give

$$\begin{aligned} |\widehat{\psi}_{\mathrm{NL}} - \psi(A)| &\leq \|w\|_\infty \sum_i \min(|a_i|, 2\lambda) \\ &\leq \|w\|_\infty (2\lambda)^{1-q} \sum_i |a_i|^q \leq h_{\mathrm{NL}}. \end{aligned}$$

This proves the uniform coverage of I_{NL}.

Because $\|L\|_F = 1$, $\langle L, Z\rangle - \psi(A) \sim N(0, \tau^2)$, so I_G has exact coverage. For any constrained unfolding, the loading matrix has singular values $|w_i|$: the product vectors on each side form orthonormal families. Lemma 4 thus gives $|\psi(A)| \leq M(\sum_{i \leq r} w_{(i)}^2)^{1/2}$. Choosing the smallest cap proves $|\psi(A)| \leq h_0$, so I_0 has coverage one. The selected index in (13) is deterministic, hence the selected interval retains its individual coverage without a multiple-selection correction. Its length is exactly (15). □

A.6 Proof of Theorem 3

Roadmap. Force a variance cost at zero and a bias cost at signed coordinate tensors, then combine the two necessary inequalities.

Proof. Set $t = \langle W, L\rangle$. At $A = 0$ the center error is normal with mean b and standard deviation $v = \tau\|W\|_F$. For fixed h and $v > 0$, the probability that this variable belongs to $[-h, h]$ is maximized at $b = 0$. Indeed, for $b \geq 0$ its derivative is

$$v^{-1}\{\phi((h+b)/v) - \phi((h-b)/v)\} \leq 0,$$

since $|h + b| \geq |h - b|$; the function is even in b. Honesty at zero therefore implies $h \geq z\tau\|W\|_F \geq z\tau|t|$. If $v = 0$, the same last inequality is automatic.

At $A = \pm ME_i$, the mean center error is

$$b \pm M(\langle W, E_i\rangle - p^{-1/2}).$$

At least one sign has absolute mean at least $M|\langle W, E_i\rangle - p^{-1/2}|$. A symmetric interval whose radius is strictly smaller than this mean magnitude has probability below $1/2$ under a nondegenerate normal distribution: it lies strictly to one side of the median. In the degenerate case the coverage is zero. Since $1 - \alpha > 1/2$, honesty forces

$$h \geq M \max_i |\langle W, E_i\rangle - p^{-1/2}| \geq \frac{M}{\sqrt{p}}|1 - t|.$$

The last inequality follows from $t - 1 = p^{-1/2}\sum_i(\langle W, E_i\rangle - p^{-1/2})$. Put $a = M/\sqrt{p}$ and $b' = z\tau$. The two bounds imply

$$1 \leq |1 - t| + |t| \leq h/a + h/b',$$

which rearranges to (17). On the rank-one tensor ball, $h_0 = M/\sqrt{p}$, and the direct/structural choice gives the matching order. No result about arbitrary data-dependent widths is invoked. □

A.7 A mixture inequality for confidence intervals

Lemma 5 (Common-target alternatives)**.** *Let P_0 have target zero and let Q be a mixture of alternatives whose targets all equal $\delta > 0$. If C covers zero under P_0 and covers the target at every alternative with probability at least $1 - \alpha$, then*

$$\mathbb{E}_0|C| \geq \delta\{1 - 2\alpha - \mathrm{TV}(P_0, Q)\}. \tag{48}$$

If $Q \ll P_0$, then $\mathrm{TV}(P_0, Q) \leq \frac{1}{2}\sqrt{\chi^2(Q, P_0)}$.

Roadmap. Transfer coverage from the common-target mixture to the null using total variation, then intersect the two coverage events.

Proof. Mixture coverage gives $Q(\delta \in C) \geq 1 - \alpha$. By the definition of total variation, $P_0(\delta \in C) \geq 1-\alpha-\mathrm{TV}(P_0, Q)$. Intersecting with $\{0 \in C\}$ gives probability at least $1-2\alpha-\mathrm{TV}$. An interval containing both points has length at least δ, proving (48). The divergence bound follows from Cauchy–Schwarz applied to $\frac{1}{2}\mathbb{E}_0|dQ/dP_0 - 1|$. Appending an independent random seed leaves total variation unchanged, so randomized intervals are covered as well. □

For Gaussian shifts $P_B = N(\mathrm{vec}(B), \tau^2 I_D)$, direct integration of their likelihood ratios gives

$$\mathbb{E}_0\left[\frac{dP_B}{dP_0}\frac{dP_C}{dP_0}\right] = \exp\{\langle B, C\rangle/\tau^2\}. \tag{49}$$

To verify this, each ratio is $\exp\{\langle Z, B\rangle/\tau^2 - \|B\|_F^2/(2\tau^2)\}$. Multiply the ratios and use the moment generating function of $\langle Z, B + C\rangle$, whose variance under P_0 is $\tau^2\|B + C\|_F^2$. The exponent simplifies to (49).

A.8 Proof of Theorem 4

Roadmap. Construct rank-one matrix alternatives within the chosen unfolding and bound their full Gaussian likelihood second moment using an auxiliary Gaussian.

Proof. For the given bipartition, let $f_i = \bigotimes_{k\in S} e_i$ and $g_i = \bigotimes_{k\notin S} e_i$. Both families are orthonormal. Let $u = p^{-1/2}(s_1, \dots, s_p)$ with independent uniform signs, and define

$$\theta = \min(M, \tau\sqrt{p}/4), \qquad A_u = \theta \left(\sum_i u_i f_i\right) \otimes \left(\sum_j u_j g_j\right).$$

Every alternative has rank one in this unfolding, norm θ, and functional value $\delta = \theta/\sqrt{p}$. Let Q average their Gaussian laws. If v is an independent copy of u, then (49) gives

$$1 + \chi^2(Q, P_0) = \mathbb{E}_{u,v} \exp\{\theta^2 (u^\top v)^2/\tau^2\} = \mathbb{E} \exp\{a S_p^2/p\},$$

where S_p is a sum of p independent signs and $a = \theta^2/(\tau^2 p) \le 1/16$.

Introduce an independent standard normal H. Its moment generating function yields

$$\begin{aligned}\mathbb{E} \exp(a S_p^2/p) &= \mathbb{E}_H \left\{\cosh(\sqrt{2a/p}\, H)\right\}^p \\ &\le \mathbb{E}_H \exp(aH^2) = (1-2a)^{-1/2} \le \sqrt{8/7}.\end{aligned}$$

The inequality uses $\log\cosh(x) \le x^2/2$: for $x \ge 0$, its derivative $\tanh(x)$ is at most x, and both functions are even. The Gaussian-square expectation follows by integrating $\exp\{-(1-2a)h^2/2\}/\sqrt{2\pi}$; it is finite since $a < 1/2$. Nonnegative integrands justify exchanging the expectations.

Lemma 5 now gives (19). The constant is positive for $\alpha < 1/4$. On Θ_S, Lemma 4 with loading singular values all equal to $p^{-1/2}$ gives $|\psi(A)| \le M/\sqrt{p}$. Choosing between this structural interval and the direct interval attains the lower order. All calculations use the entire Gaussian experiment; the result does not arise from hiding off-diagonal observations. □

A.9 Proof of Theorem 5

Roadmap. Substitute the specified dimensions and norm scaling, then apply the two lower-bound theorems with their confidence-level restrictions.

Proof. Substitute $M = \tau\sqrt{p}$, $q = 2/K$ and $\|w\|_\infty = p^{-1/2}$ into (10). This gives

$$h_{\mathrm{NL}} = \tau J^{1-1/K} p^{1/K - 1/2} \{8 \log(2p/\alpha)\}^{(K-2)/(2K)},$$

which is the radius in (20). For fixed $K \ge 3$, its exponent $\beta = (K-2)/(2K)$ is positive and $(\log p/p)^\beta \to 0$. The cap classes contain all $\pm M E_i$ and zero, so Theorem 3 gives radius at least $z\tau/(1+z)$. For any retained constrained unfolding, the corresponding matrix-rank cap class contains Θ_S, and Theorem 4 gives expected length at zero at least $c_\alpha \tau/4$. The same comparisons under the stated growing-cap condition follow by retaining the factor $J^{1-1/K}$. □

A.10 Proof of Theorem 6

Roadmap. Mix orthogonal coordinate shifts, compute the full likelihood second moment exactly and apply the interval mixture lemma.

Proof. Let $a = \min\{M, \tau\sqrt{\log(1+p/4)}\}$ and mix uniformly over $A_j = aE_j$, $j = 1, \dots, p$. Each is in the rank-one tensor class and has target $\delta = a/\sqrt{p}$. Orthogonality of the E_j and (49) imply

$$1 + \chi^2(Q, P_0) = p^{-2}\{p \exp(a^2/\tau^2) + p(p-1)\}.$$

Thus $\chi^2(Q, P_0) = \{\exp(a^2/\tau^2) - 1\}/p \le 1/4$ and $\mathrm{TV}(P_0, Q) \le 1/4$. Lemma 5 gives (21). This construction does not establish that these alternatives are least favorable. □

A.11 Proof of Proposition 1

Roadmap. Construct sparse diagonals inside the rank-one tensor class, bound the diagonal likelihood mixture using factorial moments and balance the support size. The full-data likelihood is kept separate.

Proof. For an integer $1 \le s \le p$, let S be a uniformly chosen size-s subset of $\{1, \dots, p\}$. Define

$$A_S = b_s \left(\sum_{i \in S} e_i \right)^{\otimes K}.$$

This is a rank-one tensor, $\|A_S\|_F = b_s s^{K/2} \le M$, and every alternative has target $\delta_s = s b_s/\sqrt{p}$. Its observed diagonal has mean $b_s \mathbf{1}_S$. Let Q_d be the uniform mixture of these diagonal Gaussian laws, and let $P_{0,d}$ be the zero-mean diagonal law. Applying the Gaussian shift calculation (49) in dimension p gives

$$1 + \chi^2(Q_d, P_{0,d}) = \mathbb{E}_{S,T} \exp\{b_s^2 |S \cap T|/\tau^2\}, \tag{50}$$

where S, T are independent uniform size-s subsets. This is the sparse-mixture mechanism used in classical sequence lower bounds [Collier et al., 2017]; the following counting argument records the needed bound explicitly.

Conditional on S, $H = |S \cap T|$ is hypergeometric with population size p, success count s and sample size s. Write $(x)_j = x(x-1)\cdots(x-j+1)$, with $(x)_0 = 1$. Counting the j-element subsets of S contained in T yields

$$\mathbb{E}\binom{H}{j} = \binom{s}{j} \frac{(s)_j}{(p)_j} \le \binom{s}{j} (s/p)^j, \qquad 0 \le j \le s.$$

For $j \ge 1$, the inequality follows factor by factor from $(s-\ell)/(p-\ell) \le s/p$ for $0 \le \ell < j$, since $s \le p$. Set $t = \exp(b_s^2/\tau^2) - 1 \ge 0$. Expanding $(1+t)^H$ and using the preceding factorial moments gives

$$\begin{aligned} \mathbb{E} \exp(b_s^2 H/\tau^2) &= \sum_{j=0}^{s} t^j \mathbb{E}\binom{H}{j} \\ &\le (1 + st/p)^s \le \exp(s^2 t/p) \le e^{1/4}. \end{aligned}$$

The last inequality uses the definition of b_s. Equation (50) and Cauchy–Schwarz therefore give

$$\mathrm{TV}(Q_d, P_{0,d}) \le \frac{1}{2}\sqrt{e^{1/4} - 1}.$$

Lemma 5, now applied to the diagonal experiment, proves (23). The constant is positive because $\alpha < 1/4$ and $\frac{1}{2}\sqrt{e^{1/4}-1} < 1/2$. The argument also covers independent randomization.

To prove (24), let $x_p = (p/\log p)^{1/K}$ and $s_p = \lfloor x_p \rfloor$. For sufficiently large p, $x_p/2 \le s_p \le x_p$ and $s_p \ge 1$. When $M = \tau\sqrt{p}$, the norm term in b_{s_p} is at least $\tau\sqrt{\log p}$. The noise term is at most this quantity, since $1 + p/(4s_p^2) \le 1 + p/4 \le p$ for $p \ge 2$. Thus the minimum equals the noise term. Since $K > 2$ and $s_p \le p^{1/K}$ eventually,

$$\log\{1 + p/(4s_p^2)\} \ge \log\{1 + p^{1-2/K}/4\} \ge \tfrac{1}{2}(1 - 2/K)\log p$$

for sufficiently large p. Consequently,

$$\frac{s_p b_{s_p}}{\sqrt{p}} \geq \frac{1}{2}\sqrt{\frac{1-2/K}{2}}\,\tau\left(\frac{\log p}{p}\right)^{(K-2)/(2K)}.$$

Theorem 2 with $J = 1$ gives an upper bound of the same order and deterministic length. Since the worst-case criterion is at least the criterion at zero, both rates follow. The fixed-cap extension follows by class inclusion for the lower bound and a constant $J^{1-1/K}$ for the upper bound.

For completeness, the corresponding *full-tensor* mixture would have $\langle A_S, A_T\rangle = b_s^2|S\cap T|^K$ in (49). Its likelihood second moment is not (50), and the preceding bound does not apply to it. The proposition is restricted to diagonal observations throughout. □

B Proofs of the full-data refinements

Lemma 6 (Finite product nets and injective norm)**.** *The nets used in* (25) *exist and satisfy* $\rho(B) \leq 2N_S(B)$ *for any tensor on* S^K*. Furthermore, a tensor* B *in either rank-cap class satisfies* $\|B\|_F \leq \sqrt{J}\,\rho(B)$*.*

Roadmap. Use a volume bound and product telescoping, then apply the canonical squared-weight identity.

Proof. A maximal ε-separated subset of the unit sphere is an ε-net. Balls of radius $\varepsilon/2$ around its points are disjoint and contained in the ball of radius $1+\varepsilon/2$. The ratio of their Euclidean volumes bounds the cardinality by $(1+2/\varepsilon)^s$. Compactness gives a finite maximal set. Take $\varepsilon = 1/(2K)$.

Approximate each factor of any unit product direction by a net point. Telescoping the product one factor at a time, each contraction difference has magnitude at most $\varepsilon\rho(B)$, by multilinearity and the definition of ρ. The sum is at most $K\varepsilon\rho(B) = \rho(B)/2$. Taking the supremum gives $\rho(B) \leq N_S(B) + \rho(B)/2$.

For the second assertion use Lemma 1, writing its product terms as X_j. Then

$$\|B\|_F^2 = \sum_j \langle B, X_j\rangle \leq \rho(B)\sum_j \|X_j\|_F \leq \rho(B)\sqrt{J}\left(\sum_j \|X_j\|_F^2\right)^{1/2} = \rho(B)\sqrt{J}\,\|B\|_F.$$

The middle inequality is Cauchy–Schwarz, and the last identity is the squared-weight conclusion of Lemma 1. Divide when $B \neq 0$; the zero case is immediate. □

B.1 Proof of Theorem 7

Roadmap. A finite Gaussian union bound controls every net contraction. Test acceptance on that event bounds each ordered diagonal coordinate. Summation gives the interval and intersection gives the two expected-length guarantees.

Proof. There are at most $(ep/s)^s$ subsets of size s, since $\binom{p}{s} \leq p^s/s!$ and $\log(s!) \geq \int_1^s \log x\,dx \geq s\log s - s$. Each unit product contraction of τG has distribution $N(0,\tau^2)$. Therefore the Gaussian Chernoff bound gives

$$\begin{aligned}\mathbb{P}\{\exists S : N_S(\tau G) > \lambda^{\mathrm{scan}}_{|S|}\} &\leq \sum_{s=1}^{p} 2(ep/s)^s(1+4K)^{Ks}\exp\{-(\lambda^{\mathrm{scan}}_s)^2/(2\tau^2)\}\\ &\leq \gamma.\end{aligned}$$

Let $\mathcal{E}_\gamma$ be the complementary event. No independence between the scanned contractions is required.

On $\mathcal{E}_\gamma$, if the test accepts, the triangle inequality for the finite maximum implies $N_S(A) \le 2\lambda_s^{\mathrm{scan}}$. Lemma 6 gives $\rho(A_{S,\dots,S}) \le 4\lambda_s^{\mathrm{scan}}$. Coordinate restriction cannot increase any constrained unfolding rank, because it multiplies an unfolding by selection matrices. Theorem 1 and Lemma 6 thus imply

$$\sum_{i\in S} |A_{i,\dots,i}|^q \le J^{1-q/2}\|A_{S,\dots,S}\|_F^q \le J\,\rho(A_{S,\dots,S})^q.$$

Let $|a|_{(s)}$ be the sth largest absolute diagonal entry. Choose S to contain the s largest entries. It follows that

$$s|a|_{(s)}^q \le J(4\lambda_s^{\mathrm{scan}})^q, \qquad |a|_{(s)} \le 4J^{K/2}\lambda_s^{\mathrm{scan}} s^{-K/2}.$$

Summing these inequalities and using $|\psi(A)| \le p^{-1/2}\sum_i |a_i|$ proves the bound $|\psi(A)| \le R_\gamma$ in (26). On rejection the structural interval covers surely. Therefore C_{scan} covers on $\mathcal{E}_\gamma$ and is honest at level $1-\gamma$.

Both intersected intervals cover with probability at least $1-\alpha_0-\gamma$. On that event the intersection is nonempty and covers; replacing empty intersections by $\{0\}$ preserves this argument. Its length is always at most ℓ_*. At zero, the rejection probability is at most γ, and on acceptance the intersection has length at most $2R_\gamma$. This gives (27), without independence between the intervals.

For the specified γ, elementary logarithm bounds give $\lambda_s^{\mathrm{scan}} \le C_K\tau\sqrt{s\log(2p/\alpha)}$. For example, use $s\log(ep/s) \le s(1+\log p)$, $\log(2p/\gamma) \le 2\log(2p/\alpha)$ and $s \ge 1$. Equation (26) now proves (28); the extra term $\ell_*\gamma$ is smaller in order by Theorem 5. If $K \ge 4$, the sum is at most $1+2/(K-3)$ by integration of $x^{-(K-1)/2}$ over $[1,\infty)$. If $K=3$, it is at most $1+\log p$. Combining the upper bound for $K \ge 4$ with Theorem 6 proves (29). □

Lemma 7 (Uniform cubic moment for a weighted sign sum)**.** *For independent uniform signs ϵ_i, let $S_p = \sum_{i=1}^p i^{-2/3}\epsilon_i$. Then, uniformly in $p \ge 1$,*

$$\mathbb{P}(|S_p| > x) \le 2e^{-x^3/192} \quad (x \ge 12), \qquad \mathbb{E}\exp\{|S_p|^3/40000\} \le 20/19 + 1/100 < 5/4.$$

Roadmap. Split the weighted sign sum into a bounded head and a sub-Gaussian tail, then integrate its cubic tail probability.

Proof. For $x \ge 12$, put $m = \lfloor (x/6)^3 \rfloor$. If $m \ge p$, then $|S_p| \le 3p^{1/3} \le x/2$. Otherwise, the first m terms have absolute sum at most $3m^{1/3} \le x/2$. The remaining independent sign sum has variance proxy

$$V_m = \sum_{i>m} i^{-4/3} \le 3m^{-1/3} \le 24/x.$$

Here $m \ge (x/6)^3/2$ and $18\cdot 2^{1/3} < 24$. The product inequality $\prod_i \cosh(ti^{-2/3}) \le \exp(t^2V_m/2)$ and Chernoff optimization give a two-sided tail bound $2\exp\{-(x/2)^2/(2V_m)\} \le 2e^{-x^3/192}$.

Put $u = 1/40000$ and $Y = |S_p|^3$. Tonelli's theorem applied to $e^{uY} = 1 + \int_0^Y ue^{ut}\,dt$ yields

$$\mathbb{E}e^{uY} \le e^{1728u} + \frac{2u}{1/192-u}e^{-(1/192-u)1728} \le \frac{20}{19} + \frac{1}{100}.$$

The last step uses $1728u < 1/20$, $e^x \le (1-x)^{-1}$ for $0 \le x < 1$, and $1/192 - u \ge 1/200$. The exponential factor in the second term is at most one. These estimates are uniform in p. □

B.2 Proof of Theorem 8

Roadmap. Couple three rank-one factor signs so that the diagonal is deterministic and harmonic. Control the full likelihood mixture by the cubic moment lemma and Hölder, without assuming independence of the three inner-product sums.

Proof. For independent uniform sign vectors $u, v \in \{-1, 1\}^p$, define

$$(x_u)_i = u_i i^{-1/3}, \qquad A_{u,v} = \frac{\tau}{200} x_u \otimes x_v \otimes x_{uv},$$

where $(uv)_i = u_i v_i$. Every alternative is rank one and

$$\|A_{u,v}\|_F = \frac{\tau}{200} \left(\sum_i i^{-2/3} \right)^{3/2} \le \frac{3^{3/2}}{200} \tau \sqrt{p} \le M.$$

The sum bound follows from $1 + \int_1^p x^{-2/3}\, dx \le 3p^{1/3}$. Its diagonal is $\tau/(200i)$, so the common target is $\delta = \tau H_p/(200\sqrt{p})$.

Let Q mix the full Gaussian laws over u, v. For independent copies u', v', set $\epsilon_i = u_i u'_i$ and $\eta_i = v_i v'_i$. The full tensor inner product divided by τ^2 is $S_1 S_2 S_3/40000$, where

$$S_1 = \sum_i i^{-2/3} \epsilon_i, \quad S_2 = \sum_i i^{-2/3} \eta_i, \quad S_3 = \sum_i i^{-2/3} \epsilon_i \eta_i.$$

Each has the marginal distribution in Lemma 7. They are dependent. The likelihood identity (49), the arithmetic–geometric mean inequality and Hölder give, with $u_0 = 1/40000$,

$$\begin{aligned} 1 + \chi^2(Q, P_0) &= \mathbb{E} e^{u_0 S_1 S_2 S_3} \\ &\le \mathbb{E} e^{u_0 (|S_1|^3 + |S_2|^3 + |S_3|^3)/3} \\ &\le \prod_{j=1}^{3} \{ \mathbb{E} e^{u_0 |S_j|^3} \}^{1/3} < 5/4. \end{aligned}$$

Thus $\mathrm{TV}(Q, P_0) < 1/4$. Lemma 5 gives $\mathbb{E}_0 |C| \ge (3/4 - 2\alpha)\delta$, proving (30). Since $H_p \asymp \log p$, this also proves the claimed obstruction to the earlier zero-point conjecture. □

C Supporting refinements

These separately calibrated extensions do not change the observation model of a lower bound by implication. The simulations retain the original Gaussian procedure.

C.1 Diagonal geometry

Proposition 10 (Smaller rank coefficient)**.** *In $\sum_i |a_i|^{2/K} \le C_{\mathrm{diag}} M^{2/K}$ one may take*

$$C_{\mathrm{diag,Tu}} = \min \left\{ J_{\mathrm{Tu}}^{1-1/K}, m^{1-1/K}, (\prod_k \bar{r}_k)^{1/K} \right\}.$$

For TT caps put $\bar{r}_0 = \bar{r}_K = 1$ and $\rho_k = \min(p_k, \bar{r}_{k-1} \bar{r}_k)$; one may take

$$C_{\mathrm{diag,TT}} = \min \left\{ J_{\mathrm{TT}}^{1-1/K}, m^{1-1/K}, (\prod_k \rho_k)^{1/K} \right\}.$$

The nonlinear radius remains valid with coefficient C_{diag}.

Roadmap. Add ambient and mode-subspace bounds to the original expansion bound.

Proof. Bessel gives $\sum_i |a_i|^2 \le M^2$; Hölder in dimension m gives the second coefficient. For Tucker mode projectors P_k put $\ell_{ki} = \|P_k v_{ki}\|_2^2$. Projection leaves A unchanged, so Cauchy–Schwarz and Hölder give

$$\sum_i |a_i|^{2/K} \le M^{2/K} \sum_i \prod_k \ell_{ki}^{1/K} \le M^{2/K} \prod_k (\sum_i \ell_{ki})^{1/K} \le M^{2/K} \prod_k \bar{r}_k^{1/K}.$$

The last step uses $\sum_i v_{ki}^\top P_k v_{ki} \le \operatorname{tr} P_k$. For TT cores, fixing the two bond indices of core k expresses its physical fibers through at most $\bar{r}_{k-1}\bar{r}_k$ vectors. The mode rank is therefore at most ρ_k; the Tucker argument applies. Theorem 1 gives the first coefficient. □

For equal Tucker caps r this supplies coefficient r, instead of $r^{(K-1)^2/K}$; optimal rank dependence is not established. For unit loading vectors with frame bounds $\sum_i v_{ki} v_{ki}^\top \preceq F_k I$, the original proof instead gives coefficient $J^{1-1/K} \prod_k F_k^{1/K}$. This follows by replacing each Bessel inequality by its frame bound. The direct variance must use the actual $\|L\|_F^2$; the orthogonal structural formula is not asserted for these frames.

C.2 Exact affine benchmark

Let $Q_\alpha(\mu, v)$ be the smallest $h \ge 0$ such that $\mathbb{P}\{|N(\mu, v^2)| \le h\} \ge 1 - \alpha$, for $\mu, v \ge 0$, with $Q_\alpha(\mu, 0) = \mu$.

Proposition 11 (Structural range and fixed affine optimum)**.** *The range of ψ on either cap class under (4) is exactly $[-h_0, h_0]$. Under (16) and $0 < \alpha < 1/2$, set $a_0 = M\sqrt{s_{\mathrm{cap}}/p}$. Among all honest full-data fixed-radius affine intervals, the optimal radius is*

$$h^*_{\mathrm{aff}} = \min_{0 \le t \le 1} Q_\alpha(a_0(1-t), \tau t).$$

Center $t\langle L, Z\rangle$ attains the optimum. If $a_0 \le z_{1-\alpha}\tau$, then $h^*_{\mathrm{aff}} = a_0$.

Roadmap. Attain the structural bound, force bias and variance for any affine center, and reduce to a scalar optimization.

Proof. Let T index the s_{cap} largest weights. The tensor $M \sum_{i\in T} w_i V_i / (\sum_{i\in T} w_i^2)^{1/2}$ has norm M, all constrained ranks at most s_{cap}, and target h_0. Its signed scalar multiples attain the full interval; Lemma 4 gives the reverse inclusion.

For center $\langle W, Z\rangle + b$ put $t = \langle W, L\rangle$ and $g_i = \langle W, E_i\rangle - p^{-1/2}$. All s_{cap}-sparse diagonal tensors of norm at most M belong to the class. Opposite such tensors force an absolute mean error at least

$$|b| + M(\sum_{i=1}^{s_{\mathrm{cap}}} g_{(i)}^2)^{1/2} \ge a_0|1-t|.$$

Indeed, the top-square sum is at least s_{cap}/p times the full sum, and $\|g\|_2 \ge |\sum_i g_i|/\sqrt{p} = |t-1|$. The noise standard deviation is $\tau\|W\|_F \ge \tau|t|$. Confidence greater than one half forces h to be at least the absolute mean error, by the normal median argument. For $h \ge \mu \ge 0$ the probability $F(h; \mu, v) = \Phi((h-\mu)/v) + \Phi((h+\mu)/v) - 1$ decreases with μ and with $v > 0$. For μ, compare the normal densities. For v, its derivative is

$$-\{(h-\mu)\phi((h-\mu)/v) + (h+\mu)\phi((h+\mu)/v)\}/v^2 \le 0.$$

Thus $h \ge Q_\alpha(a_0|1-t|, \tau|t|)$; zero variance follows directly. For $t < 0$ this is at least a_0, and for $t > 1$ it is at least $z_{1-\alpha/2}\tau$. Both endpoints are achievable. On $[0, 1]$, center $t\langle L, Z\rangle$ has exactly the asserted maximum bias and variance and attains the quantile bound. Continuity, including the zero-variance limit, gives a minimum. Finally $F(h; \mu, v) \le \Phi((h-\mu)/v)$ implies $Q_\alpha(\mu, v) \ge \mu + z_{1-\alpha}v$. If $a_0 \le z_{1-\alpha}\tau$, every $t \in [0, 1]$ requires radius at least a_0, attained at $t = 0$. □

At cap one, $M = \tau\sqrt{p}$ and $\alpha = 0.05$, the exact affine length is 2τ. The scalar bounded-normal-mean mechanism is classical [Stark, 1992]; the tensor reduction is proved above.

C.3 Noise envelopes and approximate ranks

Define $\Theta^{\varepsilon_A} = \{A : \exists B \in \Theta(M, \bar{r}),\ \|A - B\|_F \le \varepsilon_A\}$ for supplied $\varepsilon_A \ge 0$.

Proposition 12 (Envelope and approximation calibration)**.** *Suppose $d_i = a_i + e_i$ and, uniformly over the asserted class, $\mathbb{P}(\max_i |e_i| \le \lambda_0) \ge 1 - \alpha$. For any measurable $\Lambda \ge \lambda_0$, the threshold center has an honest radius*

$$\|w\|_\infty C_{\mathrm{diag}} M^{2/K} (2\Lambda)^{1-2/K} + \varepsilon_A$$

on Θ^{ε_A}. The structural radius is $h_0 + \varepsilon_A$. Data-dependent selection between these two intervals is valid. A direct Gaussian alternative needs its own calibration and deterministic selection or joint error allocation.

Roadmap. Use one event for every allowed threshold and a residual Lipschitz bound.

Proof. On the envelope event, Lemma 3 holds simultaneously for all $\Lambda \ge \lambda_0$, proving the assertion at $\varepsilon_A = 0$. For fixed e, the residual map $f_e(x) = \mathrm{soft}_\Lambda(x + e) - x = e - \mathrm{clip}(x + e, [-\Lambda, \Lambda])$ is one-Lipschitz. For $a_i = \langle A, V_i \rangle$ and $b_i = \langle B, V_i \rangle$,

$$|\sum_i w_i \{f_{e_i}(a_i) - f_{e_i}(b_i)\}| \le \|w\|_2 \|a - b\|_2 \le \|A - B\|_F.$$

The last step is Bessel. Add the threshold error at B. Also $|\psi(A) - \psi(B)| \le \|L\|_F \|A - B\|_F \le \varepsilon_A$. Both intervals cover on the same event, proving the selection assertion. □

For iid Gaussian errors the exact threshold is $\tau \Phi^{-1}(\{1 + (1 - \alpha)^{1/m}\}/2)$, by multiplying the m probabilities $2\Phi(\lambda_0/\tau) - 1$. Dependent Gaussian margins with variances at most τ^2 admit $\tau \Phi^{-1}(1 - \alpha/(2m))$ by Bonferroni. Marginal bounds $\mathbb{E}e^{te_i} \le e^{t^2 v^2/2}$ give $v\sqrt{2\log(2m/\alpha)}$ by Chernoff and a union bound. Bounds $\mathbb{E}|e_i|^a \le v^a$ give $v(m/\alpha)^{1/a}$ by Markov and a union bound. Independence is unnecessary for these three union-bound calibrations.

Writing $\mathcal{D}_L A = (\langle A, V_i \rangle)_i$, the same proof gives

$$|\widehat{\psi} - \psi(A)| \le \inf_B \left\{ \|w\|_\infty C_{\mathrm{diag},B} \|B\|_F^{2/K} (2\Lambda)^{1-2/K} + \|\mathcal{D}_L(A - B)\|_2 \right\}.$$

The infimum ranges over tensors with their indicated valid budget $C_{\mathrm{diag},B}$. Its unknown value is not an adaptive confidence radius.

For Tucker approximation let P_k be any leading mode singular-space projector of dimension $\min(\bar{r}_k, p_k)$ and put $B = (\bigotimes_k P_k)A$. The lifted projections commute. The identity $I - \prod_k P_k = \sum_k (\prod_{\ell<k} P_\ell)(I - P_k)$ has summands with mutually orthogonal ranges. Their squared residual norms add, each at most $\|(I - P_k)A\|_F^2$. Thus

$$\|A - B\|_F^2 \le \sum_k \sum_{j > \bar{r}_k} \sigma_j(A_{(k)})^2, \qquad \|B\|_F \le \|A\|_F.$$

This holds without a spectral gap, including ties. The distance-class proposition also covers TT approximation.

C.4 Finite variance and bounded row corruption

Proposition 13 (Replicate median calibration)**.** *Let clean tensor replicates be independent with common mean A, and every prescribed contraction have variance at most known v^2. Allow at most c adversarial replacements of entire replicates. Coordinates within a replicate may be dependent. Choose an odd block count b with*

$$b \ge 8c, \qquad b \ge 32\log(m/\alpha), \qquad n \ge 2b, \qquad k = \lfloor n/b \rfloor.$$

Partition the first bk replicates into fixed disjoint blocks. For coordinatewise medians d_i of the block means,

$$\mathbb{P}(\max_i |d_i - a_i| > 2v/\sqrt{k}) \le \alpha.$$

Thus Proposition 12 applies. For the smallest admissible odd b, the threshold has order

$$O\{v\sqrt{c/n + \log(m/\alpha)/n}\}.$$

Roadmap. Bound clean-block errors, count corrupted blocks deterministically and concentrate the original bad-block count.

Proof. Chebyshev bounds a clean block's error probability at $2v/\sqrt{k}$ by $1/4$. For a fixed coordinate let I_j be its original clean-block error indicators. They are independent and $\sum_j \mathbb{E}I_j \le b/4$. At most c summaries change even for an adaptive adversary. If the final median errs, $\sum_j I_j \ge b/2 - c \ge 3b/8$. For a Bernoulli I, $h(t) = \log \mathbb{E}e^{t(I-\mathbb{E}I)}$ satisfies $h(0) = h'(0) = 0$ and $h''(t) \le 1/4$, the variance of a tilted Bernoulli. Hence $h(t) \le t^2/8$. Independence and Markov give

$$\mathbb{P}\{\sum_j (I_j - \mathbb{E}I_j) \ge b/8\} \le e^{-tb/8 + bt^2/8} = e^{-b/32} \quad (t = 1/2).$$

A union bound over m coordinates proves the claim. Since $k \ge n/(2b)$, the rate follows, including the fixed rounding constants in b. □

Median-of-means calibration is classical [Devroye et al., 2016]. For independent Bernoulli replacements of probability ϵ, let $c = \lceil 2n\epsilon + 2\log(1/\delta) \rceil$. Their count satisfies $\mathbb{E}2^N = (1+\epsilon)^n \le e^{n\epsilon}$, giving $\mathbb{P}(N > c) \le e^{n\epsilon - c\log 2} \le \delta$. With envelope error budget α_0, coverage is at least $1 - \alpha_0 - \delta$ when the block conditions hold. These sufficient corruption conditions are not a sharp breakdown or contamination-minimax result.

C.5 Unknown Gaussian variance

Proposition 14 (Independent residual calibration)**.** *Retain $n \ge 2$ iid Gaussian full-tensor replicates with unknown noise standard deviation $\sigma > 0$. Let $d_{ji} = \langle W_j, V_i \rangle$, $\bar{d}_i = n^{-1}\sum_j d_{ji}$, $\nu = m(n-1)$, and*

$$S^2 = \nu^{-1} \sum_i \sum_j (d_{ji} - \bar{d}_i)^2, \qquad \sigma_U = S\sqrt{\nu/\chi^2_{\nu,\delta}}, \quad 0 < \delta < \alpha.$$

The denominator is the lower δ chi-square quantile. Compute the original three intervals with $\tau_U = \sigma_U/\sqrt{n}$ and $\alpha_0 = \alpha - \delta$, choosing the smallest using only S and supplied quantities. The selected interval is honest at level $1 - \alpha$. For fixed δ and $\nu \to \infty$, $\sigma_U/\sigma \to 1$ in probability, and the nonlinear expected-length order is preserved.

Roadmap. Orthogonally separate means and residuals, then condition on a conservative scale estimate.

Proof. An orthogonal transformation in the replicate index separates each sample mean from its $n-1$ residual coordinates. Joint Gaussianity and loading orthonormality give independence of all residual and mean coordinates. Therefore $\nu S^2/\sigma^2 \sim \chi^2_\nu$, independent of the vector of means, and $\mathbb{P}(\sigma_U \ge \sigma) = 1 - \delta$. Conditional on S with $\sigma_U \ge \sigma$, both stochastic intervals have failure at most α_0 and the selected index is fixed. The structural interval covers surely. Integrating conditional failure and adding δ proves honesty.

The normalized chi-square has mean one and variance $2/\nu$. Chebyshev gives convergence to one and convergence of its fixed-δ quantile to one. Thus $\sigma_U/\sigma \to 1$ and $\mathbb{E}(\sigma_U/\sigma)^2 = \nu/\chi^2_{\nu,\delta} \to 1$. The second-moment bound gives uniform integrability for powers at most one. Since the nonlinear radius is proportional to $\tau_U^{1-2/K}$, its expected order is preserved. □

Retained residuals are essential to this construction; it is not an estimate of variance from Z alone.

C.6 Confidence levels and covariance

Proposition 15 (Calibrated lower bounds)**.** *The fixed-confidence rates of Theorems 6, 4 and Proposition 1 extend to every $0 < \alpha < 1/2$ after changing constants. If known Gaussian covariance satisfies $\Sigma \succeq s_0^2 I$ with $s_0 > 0$, all-procedure isotropic lower bounds transfer with $\tau = s_0$.*

Roadmap. Tune the mixture distance below $1 - 2\alpha$; compare covariance models by adding independent noise.

Proof. Fix $0 < \varepsilon_{\mathrm{TV}} < 1 - 2\alpha$. For the coordinate mixture use $a = \min\{M, \tau\sqrt{\log(1 + 4\varepsilon_{\mathrm{TV}}^2 p)}\}$, giving $\chi^2 \le 4\varepsilon_{\mathrm{TV}}^2$. For the unfolding mixture use $\theta = \min(M, c_{\varepsilon_{\mathrm{TV}}}\tau\sqrt{p})$ with $c_{\varepsilon_{\mathrm{TV}}}^2 = \{1 - (1 + 4\varepsilon_{\mathrm{TV}}^2)^{-2}\}/2$. Its second-moment bound is at most $1 + 4\varepsilon_{\mathrm{TV}}^2$. For the diagonal support mixture use

$$b_s = \min\{Ms^{-K/2},\ \tau\sqrt{\log[1 + \log(1 + 4\varepsilon_{\mathrm{TV}}^2)p/s^2]}\}.$$

The factorial-moment bound again gives second moment at most $1 + 4\varepsilon_{\mathrm{TV}}^2$. Total variation is at most $\varepsilon_{\mathrm{TV}}$ in each case; the interval lemma has positive factor $1 - 2\alpha - \varepsilon_{\mathrm{TV}}$. The fixed-$\alpha$ rates are unchanged.

Add independent $N(0, \Sigma - s_0^2 I)$ noise to $N(A, s_0^2 I)$. Applying any Σ-procedure to the result gives an isotropic randomized procedure with the same coverage and length distributions. The isotropic all-procedure lower bound therefore applies. □

Upper envelope calibration uses the marginal variances $\mathrm{vec}(V_i)^\top \Sigma\, \mathrm{vec}(V_i)$; the direct radius uses $z_{1-\alpha/2}\sqrt{\mathrm{vec}(L)^\top \Sigma\, \mathrm{vec}(L)}$. A lower covariance bound is needed for a universal positive noise lower scale: if $\Sigma = \tau^2(I - \mathrm{vec}(L)\,\mathrm{vec}(L)^\top)$, the target is observed without noise. Cross terms $B^\top \Sigma^{-1} C$ cannot in general be ordered by a covariance bound; the randomization proof avoids that invalid step.

D Proofs of the sharp rates and computational statements

D.1 Proof of Theorem 9

Roadmap. Embed a harmonic rank-one profile on a random support; bound its full tensor likelihood using support overlap.

Proof. For a size-m support S, assign values $j^{-1/3}$, $j = 1, \dots, m$, in increasing coordinate order and zeros elsewhere; call this vector x_S. Set $A_S = b_m x_S^{\otimes 3}$. Then

$$\|A_S\|_F = b_m \left(\sum_{j=1}^{m} j^{-2/3}\right)^{3/2} \le b_m\sqrt{27m} \le M, \qquad \psi(A_S) = b_m H_m/\sqrt{p}.$$

The sum bound follows by integrating $x^{-2/3}$. All alternatives have rank one and the same target. For $H = |S \cap T|$, Cauchy–Schwarz and the decreasing profile give

$$0 \le \langle x_S, x_T\rangle \le \left(\sum_{i \in S\cap T} x_{S,i}^2 \sum_{i \in S\cap T} x_{T,i}^2\right)^{1/2} \le \sum_{j=1}^{H} j^{-2/3} \le 3H^{1/3}.$$

At $H = 0$ every term is zero. Therefore $\langle A_S, A_T\rangle \le 27 b_m^2 H$. The full Gaussian likelihood identity and the hypergeometric bound already proved in Proposition 1 imply

$$1 + \chi^2(Q, P_0) \le \mathbb{E} e^{27 b_m^2 H/\tau^2} \le \exp\{(m^2/p)(e^{27 b_m^2/\tau^2} - 1)\} \le e^{1/4}.$$

Only the hypergeometric inequality is reused, not the diagonal likelihood. Lemma 5 proves (32).

For $p \geq 256$, $m = \lfloor p^{1/4} \rfloor \geq p^{1/4}/2$, so $H_m \geq \log m \geq (\log p)/8$. Also $p/(4m^2) \geq \sqrt{p}/4 \geq p^{1/4}$, and

$$\tfrac{1}{4} \log p \leq \log(1 + p/(4m^2)) \leq \log p \leq \sqrt{p} \leq p/m.$$

For $p \geq 4$, $\log p \leq \sqrt{p}$ follows from positivity at four and the nonnegative derivative of $\sqrt{p} - \log p$. Thus the noise term is the minimum in b_m and is at least $\tau\sqrt{\log p}/(2\sqrt{27})$. Multiplication proves the stated constant. □

D.2 Proof of Theorem 10

Roadmap. Verify measurable extrema and coverage, then bound differences of feasible tensors using doubled caps.

Proof. The parameter class is compact: its norm ball is bounded and closed, and rank constraints are closed polynomial conditions on unfolding minors. Each finite-net maximum is continuous. Hence every nonempty feasible section is compact and its functional extrema are attained.

The feasible-data domain is closed. Indeed, if $Z_j \to Z$ and B_j are feasible, compactness supplies a subsequence $B_{j_k} \to B \in \Theta$; continuity preserves every constraint. For any real u, the set of Z admitting a feasible B with $\psi(B) \geq u$ is also closed by this argument, as is the analogous set with $\psi(B) \leq u$. The extrema on the feasible-data domain, extended by zero elsewhere, are therefore Borel functions. The noise event at error level α makes A feasible, uniformly with probability at least $1 - \alpha$. On that event C_F covers the target.

For $B, B' \in \mathcal{F}(Z)$, put $\Delta = B - B'$. Matrix-rank subadditivity gives the doubled unfolding caps; coordinate restriction cannot increase them. Also $N_S(\Delta) \leq N_S(Z - B) + N_S(Z - B') \leq 2\lambda_s$, so $\rho(\Delta_S) \leq 4\lambda_s$. The diagonal bound and injective-norm comparison with J_2 give

$$\sum_{i \in S} |\Delta_{i,\ldots,i}|^{2/K} \leq J_2^{1-1/K} \|\Delta_S\|_F^{2/K} \leq J_2 \rho(\Delta_S)^{2/K}.$$

Taking the s largest diagonal entries yields $|\Delta|_{(s)} \leq 4J_2^{K/2}\lambda_s s^{-K/2}$. Summation proves $|\psi(B) - \psi(B')| \leq W_\alpha$. Apply it to the attained extrema; the structural range gives the other bound $2h_0$. Empty sections have length zero.

For fixed confidence, $\lambda_s \leq C_{K,\alpha}\tau\sqrt{s \log p}$ for large p. The sum of $s^{-(K-1)/2}$ is bounded for $K \geq 4$ and at most $1 + \log p$ for $K = 3$. This proves the worst-case upper bounds. Theorem 6 matches the higher-order bound and Theorem 9 matches the cubic bound, already at zero and within the rank-one subclass. Since $\mathcal{L}_0 \leq \mathcal{L}_{\max}$, both rates follow. □

D.3 Proof of Proposition 2

Roadmap. Use a smaller failure budget and bounded target range for the upper risk; reduce estimation to common-target testing for the lower risk.

Proof. For $p \geq 2$, $\alpha_p = p^{-2}$ still gives $W_{\alpha_p} = O(r_K(p))$. The midpoint is in $[-h_0, h_0]$, including the empty-set convention. On coverage its error is at most $W_{\alpha_p}/2$; otherwise the error is at most $2h_0$. Thus

$$\sup_A \mathbb{E}_A(\widehat{\psi} - \psi(A))^2 \leq W_{\alpha_p}^2/4 + 4h_0^2 p^{-2} = O(r_K(p)^2),$$

since $h_0 = \tau\sqrt{s_{\text{cap}}}$. For a common-target prior with target δ, test whether an estimator is at least $\delta/2$. If its worst-case squared risk is R, Markov bounds the total null-plus-mixture testing error by $8R/\delta^2$. Total variation bounds this error below by $1 - \mathrm{TV}(P_0, Q)$. Hence $R \geq \delta^2(1 - \mathrm{TV})/8$. The coordinate and sparse harmonic mixtures have bounded total variation and targets of the required respective orders. □

D.4 Proof of Proposition 3

Roadmap. Combine leading-entry scan bounds with a quasi-norm tail bound and an explicit grid-net construction.

Proof. On acceptance and the common noise event, $|a|_{(s)} \le 4J^{K/2}\lambda_s s^{-K/2}$ for $s \le t$. The remaining magnitudes are at most a_t, so

$$\sum_{s>t} |a|_{(s)} \le a_t^{1-q} \sum_{s>t} |a|_{(s)}^q \le a_t^{1-q} B_q.$$

The tail vanishes for $t = p$. This proves the radius. On rejection the structural interval covers surely. Intersection gives honesty by error allocation and length at most ℓ_*. At zero rejection has probability at most γ, giving the stated expectation bound.

For the rate, substitute $\lambda_s \lesssim \tau\sqrt{s \log p}$ and $B_q = J^{1-1/K}\tau^{2/K}p^{1/K}$. The tail becomes a constant times $\tau p^{1/K-1/2}(\log p)^{(1-2/K)/2}t^{-(K-1)(1-2/K)/2}$, as asserted.

For a computable net, split each coordinate of $[-1,1]^s$ into $N = \lceil 4K\sqrt{s} \rceil$ equal intervals and normalize every nonzero product-grid vector. Rounding a unit vector incurs error at most $\sqrt{s}/N \le 1/(4K)$. The rounded vector is nonzero, and normalization adds at most the original error, giving a $1/(2K)$-net. Its size is at most $(N+1)^s \le (3+4K\sqrt{s})^s$. Use log cardinality $Ks\log(3+4K\sqrt{s})$ in the Gaussian union bound. Since $s \le p$, this still gives $\lambda_s \lesssim_K \tau\sqrt{s\log p}$. Each contraction costs $O(s^K)$ operations, proving the complexity bound. All completed levels cover on one event; adaptive continuation and intersection add no scan error. The separately calibrated C_* is intersected only once. □

D.5 Proof of Theorem 11

Roadmap. Expand in Hermite polynomials, bound overlap moments and control the possible mean shift of a low-degree estimator by its null variance.

Proof. Scale by τ and let h_ν be normalized multivariate probabilists' Hermite polynomials over the p^K coordinates. The generating identity $e^{tz-t^2/2} = \sum_{j\ge 0} H_j(z)t^j/j!$ gives $\mathbb{E}H_j(g)H_k(g) = j!\mathbf{1}_{\{j=k\}}$: multiply two generating functions and compare coefficients of their expectation e^{tu}. Coordinate independence gives multivariate orthonormality. The shift calculation similarly gives $\mathbb{E}h_\nu(g+a) = a^\nu/\sqrt{\nu!}$. The coefficient identities follow by differentiation at zero of Gaussian entire moment-generating functions.

For the finite support prior, the degree-d coefficient sum is

$$\sum_{|\nu|=d} \{\mathbb{E}_Q h_\nu(Z/\tau)\}^2 = \frac{1}{d!}\mathbb{E}_{S,T}\left(\frac{\langle A_S, A_T\rangle}{\tau^2}\right)^d = \frac{\lambda^{2d}}{d!s^{Kd}}\mathbb{E}H^{Kd},$$

by the multinomial theorem, where $H = |S \cap T|$. This uses the full likelihood.

Let $\mathrm{S}(m,j)$ count partitions of an m-element set into j nonempty unlabeled sets. Counting maps according to their number of image elements gives $H^m = \sum_{j=1}^m \mathrm{S}(m,j)(H)_j$. There are at most j^m labeled maps, so $\mathrm{S}(m,j) \le j^m/j!$. For $j \le s$, the hypergeometric factorial moments obey

$$\mathbb{E}(H)_j = (s)_j^2/(p)_j \le (s^2/p)^j = \mu^j,$$

using $(s-\ell)/(p-\ell) \le s/p$ and $s - \ell \le s$ factor by factor. For $j > s$ the moment is zero. Therefore

$$\mathbb{E}H^m \le \sum_{j=1}^m j^m\mu^j/j! \le m^m(e^\mu - 1) \le \mu e^\mu m^m.$$

The centered projection omits degree zero. Summing the remaining degrees,

$$\chi^2_{\le \mathsf{d}} \le \mu e^{\mu} \sum_{d=1}^{\mathsf{d}} \frac{\lambda^{2d}(Kd)^{Kd}}{d! s^{Kd}} \le \mu e^{\mu} \sum_{d=1}^{\mathsf{d}} B_{\mathsf{d}}^d \le \mu e^{\mu} B_{\mathsf{d}}/(1 - B_{\mathsf{d}}) = \eta_{\mathsf{d}}^2.$$

For a degree-d estimator with worst-case squared risk R on zero and the alternatives, $|\mathbb{E}_0 T| \le \sqrt{R}$ and $|\mathbb{E}_Q T - \delta| \le \sqrt{R}$. Orthonormal projection and Cauchy–Schwarz give

$$|\mathbb{E}_Q T - \mathbb{E}_0 T| \le \sqrt{\mathrm{Var}_0\, T} \sqrt{\chi^2_{\le \mathsf{d}}} \le \sqrt{R}\, \eta_{\mathsf{d}}.$$

Thus $\delta \le (2 + \eta_{\mathsf{d}})\sqrt{R}$, proving the risk lower bound. For the stated exponents, $\mu = O(p^{2a-1}) \to 0$ and $B_{\mathsf{d}} = O(p^{1-aK+bK}) \to 0$; floor factors remain bounded for large p. The target is $\delta = \tau s^{1-K/2}$. Since $a(K-2) < 1$, its square is polynomially larger than τ^2/p times either logarithmic factor of the unrestricted minimax risk. The alternatives have norm M and rank one in every unfolding, so belong to either positive cap class. □

D.6 Proof of Proposition 4

Roadmap. Combine mixture coverage with null coverage and Markov's inequality for length.

Proof. Mixture coverage gives $Q(\delta \notin C) \le \alpha$. If $0 \in C$ and $|C| < \delta$, the interval cannot contain δ. Consequently

$$P_0(\delta \in C) \le P_0(0 \notin C) + P_0(|C| \ge \delta) \le \alpha + \mathbb{E}_0|C|/\delta.$$

Add the errors. Polynomial-time interval evaluation and membership evaluation yield a polynomial-time test, proving the conditional implication. No unrestricted testing-hardness conclusion is deduced from the low-degree squared-risk bound. □

E Proofs for norm budgets and loading geometry

E.1 Proof of Theorem 12

Roadmap. Bound each ordered diagonal difference twice, then take the minimum before summing.

Proof. The coverage, compactness, and measurable-extrema argument of Theorem 10 applies for every finite $M > 0$. For two feasible tensors put $\Delta = B - B'$. Then $\|\Delta\|_F \le 2M$ and every constrained unfolding rank is at most twice its original cap. Writing $a_i = \Delta_{i,\dots,i}$ and $q = 2/K$, Theorem 1 gives

$$s|a|^q_{(s)} \le \sum_i |a_i|^q \le J_2^{1-1/K}(2M)^q.$$

The norm argument and the residual-scan proof give, respectively,

$$|a|_{(s)} \le P_2 M s^{-K/2}, \qquad |a|_{(s)} \le Q_2 \lambda_s(\alpha) s^{-K/2}.$$

Both inequalities hold for the same ordered entry. Summing their minimum bounds $|\psi(B) - \psi(B')|$ by (37). The structural range gives $2h_0$. Empty feasible sets produce a singleton and therefore length zero. □

Lemma 8 (A norm-sensitive harmonic sum)**.** *For $p \ge 1$, $x > 0$ and $K \ge 3$, put*

$$S_{K,p}(x) = \sum_{s=1}^{p} \min\{x s^{-K/2}, s^{-(K-1)/2}\}.$$

Then $S_{3,p}(x) \le 4 g_p(x)$, and $S_{K,p}(x) \le 3\min(x, 1)$ for $K \ge 4$. For fixed K and $0 < \alpha < 1$, $U_{K,p}(M; \alpha) \le C_{K,\bar{r},\alpha} \mathfrak{r}_K(p, M, \tau)$ uniformly in p, M, τ.

Roadmap. Split the cubic sum at $s = x^2$ and use summability in higher order.

Proof. For $K \geq 3$, $S_{K,p}(x) \leq x \sum_{s\geq 1} s^{-3/2} \leq 3x$. For $K \geq 4$ the second term gives $S_{K,p}(x) \leq \sum_{s\geq 1} s^{-3/2} \leq 3$.

For $K = 3$, $S_{3,p}(x) \leq H_p \leq \ell_p$. If $x \leq 1$, the bound $3x$ suffices for $g_p(x) = x$. If $1 < x < \sqrt{p}$, put $n = \lfloor x^2 \rfloor$. Since $n \geq x^2/2$,

$$S_{3,p}(x) \leq H_n + x \sum_{s>n} s^{-3/2} \leq 1 + 2\log x + 2x/\sqrt{n} \leq 4(1 + \log x).$$

If $x \geq \sqrt{p}$, the bound $H_p \leq 1 + \log p \leq 2(1 + \log x)$ gives the same conclusion. Together with $3x$ and ℓ_p, this proves the first statement.

The threshold satisfies

$$\lambda_s(\alpha) \leq c_{K,\alpha}\tau\sqrt{s\ell_p}, \qquad c_{K,\alpha} = \{2[2 + K\log(1 + 4K) + \log(2/\alpha)]\}^{1/2}.$$

Indeed, bound $\log(ep/s) \leq \ell_p$, $\log p \leq \ell_p$, $s \geq 1$, and $\ell_p \geq 1$ in (25). The inequality $\min(au, bv) \leq \max(a, b)\min(u, v)$ for nonnegative arguments shows

$$U_{K,p}(M; \alpha) \leq \max(P_2, Q_2 c_{K,\alpha}) \frac{\tau\sqrt{\ell_p}}{\sqrt{p}} S_{K,p}(x).$$

The established sum bounds prove the claim. □

E.2 Proof of Theorem 13

Roadmap. Combine the new diameter with a coordinate mixture and a norm-tuned harmonic support mixture.

Proof. The upper bound follows from Theorem 12 and Lemma 8. Since the upper diameter is deterministic it controls both expected-length criteria.

For the lower bound put $a = \min\{M, \tau\sqrt{\log(1 + p/4)}\}$ and mix uniformly over aE_i, $i = 1, \ldots, p$. These rank-one alternatives have target $a/\sqrt{p}$. The full Gaussian likelihood identity gives

$$\chi^2(Q, P_0) = \frac{e^{a^2/\tau^2} - 1}{p} \leq \tfrac{1}{4}, \qquad \mathrm{TV}(Q, P_0) \leq \tfrac{1}{4}.$$

The common-target interval lemma implies

$$\mathbb{E}_0|C| \geq (3/4 - 2\alpha)\frac{\min\{M, \tau\sqrt{\log(1 + p/4)}\}}{\sqrt{p}}. \tag{51}$$

For $p \geq 1$, $\log(1 + p/4) \geq \ell_p/5$. To verify this, the difference is positive at one and its derivative is $1/(p + 4) - 1/(5p) \geq 0$. Thus (51) is at least a constant times $\tau\sqrt{\ell_p}\min(x, 1)/\sqrt{p}$. This proves the lower bound for every $K \geq 4$, and for $K = 3$ when $x \leq 1$.

For the remaining cubic case assume first $p \geq 256$ and $x > 1$, and take

$$u = \min(p^{1/4}, x^2), \qquad m = \lfloor u \rfloor \geq 1.$$

Use the sparse harmonic mixture of Theorem 9, with the original general-budget coefficient b_m. Since $m \leq x^2$, $M/\sqrt{m} \geq \tau\sqrt{\ell_p}$. Also $m \leq p^{1/4}$ implies

$$\log(1 + p/(4m^2)) \geq \tfrac{1}{4}\log p \geq \ell_p/5,$$

where the first inequality holds for $p \geq 256$ and the second for $\log p \geq 4$. Hence $b_m \geq \tau\sqrt{\ell_p}/\sqrt{135}$. Furthermore $H_m \geq 1$ and $H_m \geq \log(m+1) \geq \log u$. Writing $t = \log x$, if $2t \leq (\log p)/4$ then $H_m \geq \max(1, 2t) \geq (1+t)/2$. Otherwise $H_m \geq (\log p)/4 \geq \ell_p/5$. In both cases

$$H_m \geq \tfrac{1}{5}\min(1 + \log x, \ell_p) = \tfrac{1}{5} g_p(x),$$

where $1 + \log x \leq x$ for $x \geq 1$ was used. Consequently

$$\mathbb{E}_0|C| \geq \frac{c_{\alpha,d}}{5\sqrt{135}} \mathfrak{r}_3(p, M, \tau). \tag{52}$$

All likelihoods here are for the full tensor; the support-mixture proof holds for the newly chosen m and b_m without modification.

For $p < 256$, $\ell_p < 7$ and $g_p(x) \leq 7\min(x, 1)$, so the coordinate mixture suffices with a changed absolute constant. Finally $\mathcal{L}_0 \leq \mathcal{L}_{\max}$ completes the proof. Every lower alternative has rank one; higher supplied caps enter only the upper constants. □

Lemma 9 (Residual fit and its noise tail)**.** *The estimator in* (40) *is Borel measurable. With* $T_0 = T(\tau G)$ *and* $\beta = 1/8$,

$$\mathbb{P}(T_0 > t) \leq \beta e^{-t^2/(2\tau^2)} \quad (t \geq 0), \qquad \mathbb{E}T_0^2 \leq 2\beta\tau^2.$$

Put $z_K = \sum_{s=1}^{\infty} s^{-K/2} \leq 3$. *Then*

$$|\widehat{\psi} - \psi(A)| \leq U_{K,p}(M; \beta) + \frac{Q_2 z_K}{\sqrt{p}}(T_0 + \tau/2), \qquad |\widehat{\psi} - \psi(A)| \leq 2h_0. \tag{53}$$

Roadmap. Use a Gaussian union bound with a common additive slack, then compare an approximate residual minimizer to the truth.

Proof. Each unit net contraction is one-Lipschitz in the Frobenius norm; finite maxima and the positive part preserve this property. Thus T is continuous and one-Lipschitz. Density of (B_j) implies $d(Z) = \min_{B\in\Theta} T(Z - B)$, although the estimator only uses a positive tolerance. The function d is one-Lipschitz as an infimum of functions with the same Lipschitz constant. The sets $\{Z : T(Z - B_j) \leq d(Z) + \tau\}$ are Borel, and their union is the entire observation space. Selecting the first such index is measurable. Hence so is $\widehat{\psi}$.

For $t \geq 0$, apply the same finite Gaussian union bound as for the original scan at thresholds $\lambda_s(\beta) + t$. Since

$$e^{-(\lambda_s+t)^2/(2\tau^2)} \leq e^{-\lambda_s^2/(2\tau^2)} e^{-t^2/(2\tau^2)},$$

the total probability is at most $\beta e^{-t^2/(2\tau^2)}$. Integrating the tail of T_0^2 proves its moment bound.

For $\widehat{A} = B_{j(Z)}$, comparison with A gives $T(Z - \widehat{A}) \leq d(Z) + \tau \leq T_0 + \tau$. Therefore for every support,

$$N_S(\widehat{A} - A) \leq 2\lambda_s(\beta) + 2T_0 + \tau.$$

The net comparison and doubled-rank argument bound its ordered diagonal entry by $Q_2(\lambda_s(\beta) + T_0 + \tau/2)s^{-K/2}$. It is also bounded by $P_2 M s^{-K/2}$. Use $\min(a, b + c) \leq \min(a, b) + c$, sum, and divide by $\sqrt{p}$ to obtain (53). Both tensors belong to the original cap class, so their targets lie in $[-h_0, h_0]$, giving the second bound. □

E.3 Proof of Theorem 14

Roadmap. Control residual-fit tails without an M-dependent failure event and reuse the same full-data mixtures for the lower bound.

Proof. Lemma 9 and $(a+b)^2 \le 2a^2 + 2b^2$ give

$$\sup_A \mathbb{E}_A(\widehat{\psi} - \psi(A))^2 \le \min\left\{4h_0^2, 2U_{K,p}(M;\beta)^2 + (8\beta+1)Q_2^2 z_K^2 \frac{\tau^2}{p}\right\}. \tag{54}$$

If $x \le 1$, use $h_0^2 \le s_{\mathrm{cap}} M^2/p$ and $\mathfrak{r}_K = M/\sqrt{p}$. If $x > 1$, then $\mathfrak{r}_K \ge \tau\sqrt{\ell_p}/\sqrt{p} \ge \tau/\sqrt{p}$; Lemma 8 absorbs both terms in the second bound into a constant times $\mathfrak{r}_K^2$.

The lower constructions in Theorem 13 have common target $\delta \ge c\mathfrak{r}_K$ and total variation bounded by a constant below one, independently of p, M, τ. For any estimator with worst-case risk R, the test $\mathbf{1}\{\widetilde{\psi} \ge \delta/2\}$ has total null-plus-mixture error at most $8R/\delta^2$ by Markov's inequality. Testing error is at least $1 - \mathrm{TV}(P_0, Q)$. Thus $R \ge \delta^2(1-\mathrm{TV})/8 \gtrsim \mathfrak{r}_K^2$. □

E.4 Proof of Theorem 15

Roadmap. Rearrange weights against ordered diagonal bounds; equalize the target of a hard mixture by a contraction in one mode.

Proof. Extend each prescribed orthonormal loading family to an orthonormal mode basis. Mode transformations preserve Gaussian isotropy and every constrained unfolding rank; restriction to the loading block cannot increase rank or Frobenius norm. Apply residual projection to that m^K block. Every pair of feasible block tensors satisfies the ordered-entry bound from Theorem 12, with $p = m$. The rearrangement inequality gives

$$|\psi(B) - \psi(B')| \le \sum_{s=1}^{m} w_s^* |\operatorname{diag}(B - B')|_{(s)} \le U_w.$$

The structural bound and the earlier measurable-extrema argument give the other assertions about this interval.

For the lower bound fix $d \le m$ and work in the coordinates corresponding to the d largest absolute weights. These weights are nonzero. Take whichever coordinate or nonnegative cubic harmonic mixture supplies the lower rate in dimension d in Theorem 13. Its common normalized diagonal target is at least $c\mathfrak{r}_K(d, M, \tau)$. In the first tensor mode apply the diagonal linear map with multipliers w_d^*/w_i on these coordinates. Its operator norm is at most one, so every alternative remains rank one with Frobenius norm at most M. Its weighted target is now the same for every alternative and equals $w_d^*\sqrt{d}$ times the original normalized target.

For each pair of alternatives, the first-mode factor inner product becomes

$$\sum_i (w_d^*/w_i)^2 u_i v_i \le \sum_i u_i v_i,$$

because the original factors are nonnegative. The other factor inner products are unchanged and nonnegative. Thus the full tensor inner product does not increase. The Gaussian likelihood second moment is no larger than that of the original hard mixture. Embedding these alternatives in the full ambient tensor adds only independent mean-zero noise coordinates and leaves the likelihood second moment unchanged. The common-target interval lemma therefore gives $\mathbb{E}_0|C| \ge c_{K,\alpha} w_d^* \sqrt{d}\, \mathfrak{r}_K(d, M, \tau)$. Maximize over the finite set of d to obtain (42). □

E.5 Proof of Proposition 5

Roadmap. Rotate to the loading block, absorb known signs, and apply the matching dimension-m theorem.

Proof. Replace each first-mode loading vector by $\epsilon_i v_{1i}$, preserving orthonormality. The loading now has equal positive weights $m^{-1/2}$. The reduced m^K Gaussian tensor has the same norm and cap constraints, so Theorems 13 and 14 give the upper bounds. For the lower bounds embed their rank-one alternatives in the loading block and set all other mean entries to zero. The extra independent noise carries no parameter information; the full likelihood second moments equal the reduced ones. Equivalently, the interval lower bound follows by taking $d = m$ in Theorem 15, and the same mixtures give the squared-risk lower bound. This includes $m = 1$ and arbitrary ambient dimensions at least m. □

F Proofs for robustness, loading transfer, and adaptation

F.1 Proof of Theorem 16

Roadmap. Identify the tail inequalities actually used by the upper proofs, then use a Gaussian subfamily for family-wise minimax lower bounds.

Proof. For every unit net direction, Chernoff applied to (43) gives $\mathbb{P}(|\langle U, E\rangle| > t) \le 2e^{-t^2/(2v^2)}$. The finite union bound proving scan coverage and the additive-slack bound proving Lemma 9 use no independence between directions. Both therefore hold with v replacing τ. All subsequent diameter and squared-error inequalities are deterministic functions of these events and moments. Orthogonal changes of coordinates preserve (43). The weighted proofs apply unchanged. The unknown-law family includes $E \sim N(0, v^2 I)$. Honesty over the family implies honesty over that subexperiment, and the worst-case risk or expected length over the family is at least its Gaussian value. The Gaussian lower bounds give the matching order. □

F.2 Proof of Theorem 17

Roadmap. Control every median by a binomial tail, obtain a common excess-tail moment, and reuse the ordered-difference argument.

Proof. For a direction at level s, each block average has variance at most σ^2/k_s. Chebyshev gives a block-error probability at most $\sigma^2/(k_s t^2)$. If the median errs by more than t, at least half its blocks err. Independence of blocks, followed by a union over their subsets, gives

$$\mathbb{P}(|d_U - \langle U, A\rangle| > \lambda_s^{\mathrm{mom}} + t) \le 2^{-b_s}(1 + t/\lambda_s^{\mathrm{mom}})^{-b_s}, \quad t \ge 0.$$

This follows by substituting $\lambda_s^{\mathrm{mom}} = 4\sigma/\sqrt{k_s}$ into $2^{b_s}\{\sigma^2/[k_s(\lambda_s^{\mathrm{mom}} + t)^2]\}^{b_s/2}$. There are at most e^{H_s} directions at level s. By the definition of b_s, $\sum_s e^{H_s} 2^{-b_s} \le \beta$. Thus all constraints cover the true tensor with probability at least $1 - \beta$.

Since $k_s \ge n/(2b_s)$, $\lambda_s^{\mathrm{mom}} \le 4\sqrt{2}\tau\sqrt{b_s}$. Define $T_0 = \max_U(|d_U - \langle U, A\rangle| - \lambda_{s(U)}^{\mathrm{mom}})_+$. For $a \ge 0$, the function $b\log(1 + a/\sqrt{b})$ is increasing for $b > 0$: its derivative is $\log(1 + u) - u/[2(1 + u)] \ge 0$, $u = a/\sqrt{b}$. Because every $b_s \ge 9$, the union bound yields

$$\mathbb{P}(T_0 > t) \le \beta(1 + t/(12\sqrt{2}\tau))^{-9}, \qquad \mathbb{E}T_0^2 \le \frac{72}{7}\beta\tau^2. \tag{55}$$

For the moment, integrate $2t$ times the tail and use $\int_0^\infty u(1 + u)^{-9} du = 1/(8 \cdot 7)$.

For two feasible tensors every directional difference is at most $2\lambda_s^{\mathrm{mom}}$. The net comparison, doubled caps and norm bound prove exactly the previous norm-sensitive diameter. Medians are measurable finite

order statistics. The extrema of their compact feasible sections are Borel by the same compact-section argument as before, now treating $(d_U)_U$ as the finite observation vector. The smallest-odd-integer rule gives $b_s \le 11 + [H_s + \log(p/\beta)]/\log 2 \le C_{K,\beta} s\ell_p$. The sum lemma therefore bounds the diameter by $C\mathfrak{r}_K$.

For estimation, F is continuous and one-Lipschitz in B. Selecting the first dense-sequence element with value at most $\inf_j F(B_j) + \tau$ is measurable. Its comparison with A gives directional differences at most $2\lambda_s^{\mathrm{mom}} + 2T_0 + \tau$. Lemma 9's deterministic argument gives

$$|\widehat{\psi} - \psi(A)| \le U^{\mathrm{mom}} + Q_2 z_K (T_0 + \tau/2)/\sqrt{p}, \qquad |\widehat{\psi} - \psi(A)| \le 2h_0.$$

Equation (55) and the small-M structural bound prove risk $O(\mathfrak{r}_K^2)$ exactly as in Theorem 14. For the lower bound restrict the replicate family to iid Gaussian errors with covariance $\sigma^2 I$. Orthogonal transformation in the replicate index separates the sample mean, whose noise level is $\sigma/\sqrt{n}$, from mean-independent residuals. Thus the full replicate likelihood second moments equal the sample-mean ones. The earlier Gaussian lower bounds apply to any procedure using all replicates, including randomized ones. Finally, H_s is increasing in s, so the replicate condition has the stated sufficient order p for fixed K, β. □

F.3 Proof of Proposition 6

Roadmap. Exhibit exact scalar recovery and a correlated direction with inflated variance.

Proof. If $|A| < v$, then $\mathrm{sign}(A + v\xi) = \xi$. The estimator $Z - v\,\mathrm{sign}(Z)$ is identically A, proving (i). The Rademacher moment-generating function obeys $\cosh(tv) \le e^{t^2 v^2/2}$. For (ii), take the unit product tensor with every entry $p^{-K/2}$. Its contraction with E is $v\xi p^{K/2}$, with variance $v^2 p^K$. Each individual entry still has variance and sub-Gaussian proxy v^2. □

F.4 Proof of Theorem 18

Roadmap. Inflate scan constraints for a bounded approximation residual, then combine two nested lower-bound subclasses.

Proof. Write $A = B_0 + R$, $B_0 \in \Theta$, $\|R\|_F \le \varepsilon_A$; compactness ensures such a decomposition exists. On the Gaussian scan event every residual contraction of $Z - B_0$ is at most $\lambda_s + \varepsilon_A$, so B_0 is feasible. Inflate its projected target interval by ε_A on both sides. If the feasible set is empty use a singleton; this cannot occur on the coverage event. Since $\|L\|_F = 1$, the enlarged interval covers $\psi(A)$. For feasible differences the diameter argument gives

$$|C| \le U_{K,p}(M;\alpha) + \{Q_2 z_K/\sqrt{p} + 2\}\varepsilon_A,$$

using $\min(a, b + c) \le \min(a, b) + c$. It also gives $|C| \le 2h_0 + 2\varepsilon_A$. Choose between this procedure and the direct Gaussian interval using their deterministic length bounds. This proves the upper order $\min\{\tau, \mathfrak{r}_K + \varepsilon_A\}$.

The class contains Θ, giving lower bound $c\mathfrak{r}_K$ at zero. It also contains $\{tL : |t| \le \varepsilon_A\}$ because $0 \in \Theta$. Comparing zero with tL for $t = \min(\varepsilon_A, \tau/2)$ gives total variation at most $\frac{1}{2}\sqrt{e^{1/4} - 1} < 1/2$ and target difference t. Hence the lower bound is also $c_\alpha \min(\varepsilon_A, \tau)$. Finally $\min(\tau, r + \varepsilon_A) \le r + \min(\tau, \varepsilon_A) \le 2\max\{r, \min(\tau, \varepsilon_A)\}$ for every $r \ge 0$. This establishes both criteria up to fixed constants. □

F.5 Proof of Proposition 7

Roadmap. Embed sparse diagonal tensors of rank s, then compare with the full norm ball.

Proof. For a uniform size-s support S, set $A_S = b\sum_{i\in S} E_i$ with

$$b = \min\{M/\sqrt{s}, \tau\sqrt{\log(1 + p/(4s^2))}\}.$$

All constrained ranks are s, its norm is $b\sqrt{s} \le M$, and its target is $sb/\sqrt{p}$. The full tensor inner product is $b^2|S\cap T|$; the hypergeometric factorial-moment inequality gives $1+\chi^2(Q, P_0) \le \exp\{(s^2/p)(e^{b^2/\tau^2} - 1)\} \le e^{1/4}$. The common-target lemma proves the lower bound for every $s \le r_*$. Maximize over s. If there are no rank restrictions, the line $\{tL : |t| \le M\}$ belongs to the Frobenius ball. The two-point argument with $t = \min(M, \tau/2)$ gives lower order $\min(M, \tau)$. The direct interval and $[-M, M]$, chosen deterministically by radius, give the matching upper bound for every mean in that ball. No asymptotic fixed-cap argument applies to this enlarged class. □

F.6 Proof of Proposition 8

Roadmap. Add an independently calibrated error event, without assuming stochastic independence.

Proof. On the intersection of $\langle L_0, A\rangle \in C_0$ and $|\langle R, E\rangle| \le q_R$, adding the second target contribution gives the asserted interval. A union bound controls failure by $\alpha_0 + \alpha_1$. The length is the displayed sum. Chernoff or the Gaussian distribution of $\langle R, E\rangle$ supplies the calibrations. For a prescribed partition, repeat the argument over blocks using the sum of their allocated failures; their centers may be dependent. Selecting among complete constructions from deterministic diameter bounds uses no data and retains its own coverage guarantee. Applying the result with $L_0 = 0$ also covers every L, although it then reduces to the direct interval. □

F.7 Proof of Theorem 19

Roadmap. Match elementary constructive upper bounds with the existing small-budget lower bound and a product-direction two-point test.

Proof. In (i), the rate formula has $x \le 1$ and is $M/\sqrt{p}$. The structural radius is $M\sqrt{s_{\mathrm{cap}}/p}$, with fixed cap factor. Theorem 13 gives the matching lower bound. For (ii), choose the sign of U so that $\langle L, U\rangle \ge c\|L\|_F$ and compare zero with tU, $t = \min(M, \tau/2)$. The full likelihood second moment is $e^{t^2/\tau^2} \le e^{1/4}$ and the target difference is at least $ct\|L\|_F$. The common-target lemma gives the lower order at zero. Cauchy–Schwarz supplies structural radius $M\|L\|_F$, while the direct Gaussian radius is $z_{1-\alpha/2}\tau\|L\|_F$. Their deterministic selection controls worst-case length and is polynomial in the number of observed coordinates. The cap class is used only for the lower inclusion; fixed rank caps are unnecessary here. □

F.8 Proof of Proposition 9

Roadmap. Find a one-dimensional Gaussian submodel inside every such convex relaxation.

Proof. The average of ME_i, $i = 1, \dots, p$, is $(M/\sqrt{p})L$. Both signs and zero belong to the convex hull, hence so does the asserted line segment. Compare zero with tL for $t = \min(M/\sqrt{p}, \tau/2)$ and apply the same two-point calculation. At $M = \tau\sqrt{p}$ it forces order τ. The previous minimax rates divided by τ tend to zero for each fixed $K \ge 3$. This argument changes the honesty class and makes no complexity assumption. □

F.9 Proof of Theorem 20

Roadmap. Shift the full-data hard mixtures into an unused orthogonal block.

Proof. Embed the dimension-m rank-one alternatives from Theorem 13, with norm budget M_0, in coordinates $\{q+1,\dots,p\}^K$. Denote them by B. Their supports are disjoint from A_0, so $\|A_0+B\|_F^2 = \|A_0\|_F^2 + \|B\|_F^2 \le M^2$. Unfolding rank subadditivity shows A_0+B is in the outer cap class. Translation of Gaussian likelihoods by A_0 leaves their second moments unchanged. All alternatives have a common increment in the normalized p-diagonal target equal to $\sqrt{m/p}$ times their dimension-m target. Their total variation from the law at A_0 is bounded by a fixed constant less than $1-2\alpha$. Apply the common-target lemma with baseline target $\psi(A_0)$, or translate the interval by that fixed number. This proves the lower bound. □

F.10 Proof of Theorem 21

Roadmap. Recover ranks on one spectral event, bound a second-order projection residual, and condition on the independent pilot.

Proof. For each unfolding use 1/4-nets of its two unit spheres, of total cardinality at most $9^{p_k+d_k}$. Product telescoping bounds its operator norm by twice the maximum net contraction. Chernoff and a union bound over the K unfoldings give an event $\mathcal{E}$ of probability at least $1-\delta$ on which every pilot error operator norm is at most η_k. No independence between unfoldings is used. The singular-value perturbation inequality $|\sigma_j(Z_{(k)}^{(1)}) - \sigma_j(A_{(k)})| \le \eta_k$ follows from the variational characterization of singular values: for every unit vector the two image norms differ by at most η_k, and taking the relevant min/max over subspaces preserves this bound. Thus zero singular values produce observed values at most η_k, while nonzero values exceeding $2\eta_k$ produce observed values greater than η_k. All selected ranks are exact on $\mathcal{E}$. If any is zero then $A=0$ and the singleton covers.

For a positive selected rank r_k, write $A_{(k)} = U_k S_k V_k^\top$ using its true nonzero singular values. Because Q_k is the leading observed rank-r_k projector,

$$\|(I-Q_k)A_{(k)}\|_{\mathrm{op}} \le \sigma_{r_k+1}(Z_{(k)}^{(1)}) + \eta_k \le 2\eta_k.$$

For full row rank the residual is zero, and the same inequality holds. Right multiplication by $V_k S_k^{-1}$ gives

$$\|(I-Q_k)U_k\|_{\mathrm{op}} \le \frac{2\eta_k}{\sigma_{r_k}(A_{(k)})} \le \frac{2\eta_k}{\sigma_{r_k}(Z_{(k)}^{(1)}) - \eta_k}.$$

Clipping at one yields d_k^*. This calculation uses the gap only on the stated separated class, with a strictly positive observed denominator.

Expand the identity into the 2^K mutually orthogonal products of Q_k and $I-Q_k$. The projector T retains exactly the terms with at most one complement. Each term discarded by T appears in at least one product $(I-Q_k)(I-Q_\ell)$. Orthogonality therefore implies

$$\|(I-T)A\|_F^2 \le \sum_{k<\ell} \|(I-Q_k)(I-Q_\ell)A\|_F^2 \le M^2 \sum_{k<\ell} (d_k^* d_\ell^*)^2.$$

The last inequality uses $A = (\bigotimes_k U_k)G$ with $\|G\|_F = \|A\|_F$ and the operator norm of a product of mode maps. Also $\|(I-T)A\|_F \le M$. Since T is orthogonal, Cauchy–Schwarz bounds $|\langle L, (I-T)A\rangle|$ by B_L.

Condition on the pilot. The vector TL is now deterministic, and independence makes the second error contraction sub-Gaussian with proxy $v_2\|TL\|_F$. Conditional on any pilot in $\mathcal{E}$, the projected interval has error at most $\alpha-\delta$. The direct alternative has the same conditional failure bound and the structural alternative never fails. A choice determined by the pilot thus also has conditional failure at

most $\alpha - \delta$. Intersecting with a surely valid structural interval preserves coverage; empty intersections may be replaced by a singleton. Integrating and adding $\mathbb{P}(\mathcal{E}^c) \le \delta$ proves honesty. SVDs, finite sums, and tensor matrix products give the stated arithmetic complexity.

For Gaussian artificial splitting the two centered errors are jointly Gaussian with zero cross covariance, each of covariance $2\tau^2 I$, hence independent. Zero covariance would not suffice for non-Gaussian errors. No minimax length conclusion on the original unseparated class follows from this proof. □

G Optimization contracts, loadings and cap-valid inference

Observe the entire tensor $Z = A + \tau G$, with iid standard Gaussian entries, known $\tau > 0$, and $A \in \Theta(M, \bar{r})$, a Tucker or TT Frobenius ball. The functional is $\psi_L(A) = \langle L, A \rangle$; normalization of L is unnecessary. Write $J = J(\bar{r})$ for the existing canonical expansion bound: $A = \sum_{j \le J} a_j U_j$, each U_j a unit product tensor, and $\sum_j a_j^2 = \|A\|_F^2$. Zero terms can be appended. Assume positive caps.

For reference, the previously established equal/flat-loading rate is $\mathfrak{r}_K(m, M, \tau) = \tau\sqrt{\ell_m/m}\, g_K(x)$, where $\ell_m = \log(em)$, $x = M/(\tau\sqrt{\ell_m})$, and

$$g_K(x) = \begin{cases} \min\{x, 1 + \max(0, \log x), \ell_m\}, & K = 3, \\ \min\{x, 1\}, & K \ge 4. \end{cases}$$

The original equal diagonal has $m = p$ and weights $1/\sqrt{p}$. Length criteria below mean the infimum expected length at zero ($\mathcal{L}_0$), or the infimum worst-case expected length ($\mathcal{L}_{\max}$), over intervals honest on the stated parameter class.

G.1 C1. A quantitative contract for approximate minimax optimization

Use every support and every direction in the calibrated finite product nets. For their unit directions U and thresholds $\lambda_{s(U)}$, put

$$F_Z(B) = \max\{0, \max_U(|\langle U, Z - B\rangle| - \lambda_{s(U)})\}.$$

This function is 1-Lipschitz in Frobenius norm. Suppose an algorithm returns $\widehat{B} \in \Theta$ and numbers $l \le \inf_{B \in \Theta} F_Z(B) \le F_Z(\widehat{B}) = u$, with $u - l \le \varepsilon$. On the simultaneous noise event, the true A is feasible, so $F_Z(A) = 0$ and $u \le \varepsilon$. Consequently, for every support size s,

$$N_S(\widehat{B} - A) \le 2\lambda_s + \varepsilon.$$

With $J_2 = J(2\bar{r})$, the preceding ordered-diagonal proof gives

$$|\widehat{B}_{i,\dots,i} - A_{i,\dots,i}|_{(s)} \le \min\{2J_2^{(K-1)/2} M, 4J_2^{K/2}(\lambda_s + \varepsilon/2)\} s^{-K/2}.$$

For known orthogonal loading weights w_i, sorted by absolute value, the sum of the right side against w_s^* is an honest radius about $\psi_L(\widehat{B})$. It may be intersected with the structural range. Choosing $\varepsilon \le c\tau$ retains the existing all-budget minimax orders for equal/flat loadings. Indeed every λ_s is bounded below by a positive constant times τ, so the added term changes the existing minimum by only a fixed factor, including in the norm-limited regime.

For squared error set the scan error budget to 1/8. If the optimizer guarantees $\varepsilon \le c\tau$ for *every observation*, the same excess tail proof as before applies: $F_Z(\widehat{B}) \le F_Z(A) + \varepsilon$ and $N_S(\widehat{B} - A) \le 2\lambda_s + 2F_Z(A) + \varepsilon$. Thus the existing all-budget MSE bound follows. A budget-limited run with a larger reported gap does not establish that MSE guarantee.

Endpoint version. Let $\mathcal{F} = \{B \in \Theta : F_Z(B) = 0\}$. If certified outer endpoints enclose $\psi_L(\mathcal{F})$, they define an honest interval, even at an arbitrary computation stop. If feasible witnesses establish each

outer endpoint within ε of the exact endpoint, its length exceeds the ideal projection length by at most 2ε. An inner optimization solution alone does not give an honest interval. On a certified empty feasible set use $\{0\}$, as before.

Proof. Lipschitzness follows from $\|U\|_F = 1$, the reverse triangle inequality, and stability of finite maxima. The objective-gap implication uses $F_Z \geq 0$. Rank caps double under subtraction; the norm budget doubles as well. The ordered bound is exactly the earlier net comparison and diagonal quasi-norm inequality with $2\lambda_s$ replaced by $2\lambda_s + \varepsilon$. The Gaussian union event has probability at least $1 - \alpha$. Endpoint containment proves coverage on the same event. For the length assertion, use an inner feasible witness at each end. The squared-risk argument invokes the already supplied excess-tail integration, with only its deterministic tolerance constant changed. No algorithmic complexity conclusion follows from this contract.

G.2 C2. An implemented global solver on the binary rank-one class

For $p = 2$ and any fixed order $K \geq 3$, every rank-one tensor of norm at most M has a representation

$$B(a, \theta) = a \bigotimes_{k=1}^{K} (\cos\theta_k, \sin\theta_k), \quad a \in [-M, M], \quad \theta_k \in [0, \pi].$$

Each vector is represented up to sign on a semicircle; the product of signs is absorbed into a. The zero tensor is included. There is no grid restriction on the parameter space.

The code includes both singleton supports and the full support. For the latter it uses $n_K = \lceil \pi K \rceil$ equally spaced angles in $[0, \pi)$ per mode. Every unit vector is within Euclidean distance $1/(2K)$ of a net point up to sign: its angular discrepancy modulo π is at most $\pi/(2n_K)$. Absolute contractions remove the signs. The same telescoping proof therefore gives $\rho \leq 2N$. There are $\mathsf{d}_1 = 2$ singleton directions and $\mathsf{d}_2 = n_K^K$ full-support directions. Set

$$\lambda_s = \tau\sqrt{2\log(4\mathsf{d}_s/\alpha)}, \qquad s = 1, 2.$$

The sum of the two Gaussian union bounds is at most α.

For each parameter box, exact real-arithmetic ranges of cosine and products enclose every scanned contraction and tensor entry. If their contraction range is $[b_U^-, b_U^+]$, then

$$l_{\text{box}} = \max\{0, \max_U [\text{dist}(\langle U, Z\rangle, [b_U^-, b_U^+]) - \lambda_{s(U)}]\}$$

is a lower bound for F_Z throughout the box. Centers provide actual rank-one upper witnesses. A largest-scaled-side split covers the parent with two children. Branch and bound retains global lower and upper bounds; the objective run ends at its requested gap or reports a budget stop. It implements the ideal residual criterion itself on this class.

For confidence projection reject a box only when its objective lower bound is positive. Loading-entry interval arithmetic bounds every surviving box's target; their hull contains the ideal confidence projection. Feasible amplitude intervals at fixed central factors supply inner witnesses. An endpoint gap is reported only when such witnesses exist. Endpoints need not reach a requested tolerance within a finite node budget, especially near singular feasibility boundaries. This is recorded, not treated as a successful exact optimization.

The code supports arbitrary loadings in the binary tensor, but caps above one and mode dimensions above two are not implemented. Exhaustive scans and global subdivision are exponential. Tests check range containment and global certificates against independent finite calculations. Floating point evaluations include conservative guards; these are numerical certificates under the derived inequalities, not formal directed-rounding interval proofs. The real-arithmetic method, rather than any particular floating point tolerance, is the mathematical result.

G.3 L1. Sharp small-budget rates for every loading tensor

Define the product overlap

$$\rho(L) = \sup_{\|u_k\|_2=1} |\langle L, \bigotimes_k u_k \rangle|.$$

For fixed positive caps, fixed $0 < \alpha < 1/4$, arbitrary mode dimensions, and *every* known loading tensor L, if $0 < M \le \tau$, then

$$\mathcal{L}_0(\Theta; L) \asymp \mathcal{L}_{\max}(\Theta; L) \asymp M\rho(L), \qquad \inf_{\widehat{\psi}} \sup_{\Theta} \mathbb{E}_A(\widehat{\psi} - \psi_L(A))^2 \asymp M^2\rho(L)^2.$$

Constants depend only on the fixed caps/order and confidence, not on L or the ambient dimensions. The zero loading has identically zero risk and length. More generally, for all $M, \tau > 0$,

$$c_\alpha \rho(L) \min(M, \tau) \le \mathcal{L}_0 \le \mathcal{L}_{\max} \le 2 \min\{\sqrt{J} M\rho(L),\ z_{1-\alpha/2}\tau\|L\|_F\}.$$

Proof. The canonical expansion and Cauchy–Schwarz give $|\psi_L(A)| \le \rho(L) \sum_j |a_j| \le \sqrt{J} M\rho(L)$. This gives a deterministic interval and the risk of estimator zero. The direct Gaussian interval gives the other upper bound; choose between the two using known quantities. Compactness gives a maximizing product U; choose its sign so $\langle L, U\rangle = \rho(L)$. Both 0 and tU belong to every positive cap class, for $t = \min(M, \tau)/2$. Their Gaussian total variation is $2\Phi(t/(2\tau)) - 1 \le 2\Phi(1/4) - 1 < 1/4$. Honesty at both parameters implies $\mathbb{E}_0|C| \ge t\rho(L)(1 - 2\alpha - \mathrm{TV})$. The two-point squared-loss lower bound is a fixed positive multiple of $t^2\rho(L)^2$: classify an estimate by its nearest target. A classification error entails squared error at least $t^2\rho(L)^2/4$, while the sum of the two testing errors is at least $1 - \mathrm{TV}$. Therefore the larger squared risk is at least $t^2\rho(L)^2(1 - \mathrm{TV})/8$. For $M \le \tau$ the bounds match. This proof is a direct specialization of standard Gaussian two-point reasoning, not a novel general inference principle.

Limits. Computing $\rho(L)$ for a general high-order loading is itself a global optimization problem. A certified upper bound can replace it in the interval; a certified product witness supplies a lower bound. Without a constant ratio between them, an efficient sharp algorithm has not been obtained. This theorem covers all loading tensors in the stated budget regime; it does not characterize their larger-budget rates. The flat loading small-budget theorem extends to $M \lesssim \tau\sqrt{\log m}$, a larger regime than the arbitrary-loading result here.

G.4 L2. An optimized common-target lower bound for unequal weights

For an orthogonal product loading with decreasing nonzero absolute weights $w_1 \ge \cdots \ge w_m > 0$, choose $d \le m$ and $0 < \Delta \le Mw_d$. For signed original weights $\widetilde{w}_i$, set $A_i = (\Delta/\widetilde{w}_i)U_i$, $1 \le i \le d$. All alternatives have target Δ, satisfy the norm budget, and have rank one. They are orthogonal as full tensors. Let

$$v_i = \exp\{\Delta^2/(\tau^2 w_i^2)\} - 1, \quad H_d(\Delta) = \sum_{i=1}^{d} v_i^{-1}, \quad \pi_i = v_i^{-1}/H_d(\Delta).$$

The optimal chi-square divergence among mixtures on these alternatives is

$$\chi^2(Q, P_0) = \min_{\pi \in \mathcal{S}_d} \sum_i \pi_i^2 v_i = H_d(\Delta)^{-1}.$$

Thus every interval honest on the full tensor class satisfies

$$\mathbb{E}_0|C| \ge \Delta \left[1 - 2\alpha - \frac{1}{2\sqrt{H_d(\Delta)}}\right]_+.$$

Proof. The Gaussian likelihood identity gives overlap 1 for distinct orthogonal alternatives and $1+v_i$ for equal alternatives. Subtracting 1 leaves $\sum \pi_i^2 v_i$. Cauchy–Schwarz bounds this below by $(\sum v_i^{-1})^{-1}$ and the displayed weights attain equality. Apply TV $\le \sqrt{\chi^2}/2$ and the common-target interval lemma. This uses the full observation; there is no diagonal-data restriction.

`weighted_axis_lower` enumerates d and uses scalar root finding for the largest $\Delta \le M w_d$ with $\chi^2 \le 1/4$. Every reported lower bound uses the actual recomputed divergence and a feasible prior. This optimizes the prior at each chosen target within the axis family. It is not an optimization over all tensor alternatives. For equal weights it recovers the one-spike logarithmic obstruction. It cannot recover the cubic harmonic obstruction on its own and is not claimed sharp for all weights.

G.5 A1. Unseparated rate adaptation across positive fixed caps

Fix an outer positive Tucker or TT cap vector $\bar{r}$. Let $\mathcal{R}$ contain any collection of positive cap vectors componentwise below it. For equal or flat orthogonal loadings, the *same* full-data outer-class interval satisfies

$$\sup_{A \in \Theta(M,r)} \mathbb{E}_A |C_{\bar{r}}| \le C_{K,\bar{r},\alpha}\, \mathfrak{r}_K(m, M, \tau), \qquad r \in \mathcal{R}.$$

Every smaller class contains the rank-one ball, whose lower bound is $c_{K,\alpha}\mathfrak{r}_K$. Therefore the interval is simultaneously minimax in rate over every such class, uniformly in m, M, τ, without knowing the actual ranks or imposing singular-value separation. The analogous MSE conclusion holds for the outer-class ideal residual fit.

Proof. Restrict the existing outer supremum upper bound to each subset and apply the same rank-one lower bound inside each smaller class. No selection step or new probability argument is required.

This observation resolves the fixed-positive-cap, rate-only statistical adaptation question. It does not give optimal rank-dependent constants, growing-cap rates, computation, exact-rank selection, or pointwise oracle lengths. Rank zero must be excluded: its known-model oracle interval has length zero, whereas outer-class honesty forces positive length at zero. The preceding ledgers used the word adaptation broadly; these distinct objectives must be separated.

G.6 A2. A computable projection certificate without separation

Let $Z^{(1)} = A + E^{(1)}$ and $Z^{(2)} = A + E^{(2)}$ have independent iid Gaussian errors of known scales τ_1, τ_2. Only $\|A\|_F \le M$ is assumed here; supplied rank caps are unnecessary. Let $0 < \delta < \alpha$. Use the simultaneous unfolding operator envelopes from the previous stage:

$$\eta_k = 2\tau_1 \sqrt{2\{(p_k + \prod_{l \ne k} p_l)\log 9 + \log(2K/\delta)\}}.$$

On one event of probability at least $1-\delta$ all $\|E^{(1)}_{(k)}\|_{op} \le \eta_k$. For any pilot-chosen ranks r_k, let Q_k project onto the first r_k left singular vectors of $Z^{(1)}_{(k)}$. Set $d_k = \sigma_{r_k+1}(Z^{(1)}_{(k)}) + \eta_k$ if $r_k < p_k$, and $d_k = 0$ if $Q_k = I$. Singular values beyond the matrix rank are zero. For $P = \bigotimes_k Q_k$, decompose

$$R = (I-P)L = \sum_{k=1}^{K} R_k, \quad R_k = \left[\bigotimes_{l<k} Q_l\right] \otimes (I - Q_k) \otimes \left[\bigotimes_{l>k} I\right] L.$$

Then the observable bound

$$B = \min\left\{ M\|R\|_F,\ \sum_k d_k \|(R_k)_{(k)}\|_* \right\}$$

satisfies $|\langle R, A\rangle| \le B$ on that event. The interval centered at $\langle PL, Z^{(2)}\rangle$, with radius

$$B + z_{1-(\alpha-\delta)/2}\tau_2\|PL\|_F,$$

is honest over the whole norm ball. Any finite set of pilot-selected projection choices can be compared by their radii on this same event; choose the smallest before using validation data. Include $P = I$ for the direct interval and $P = 0$ for the structural interval. Intersect with $[-M\|L\|_F, M\|L\|_F]$ and use $\{0\}$ for an empty intersection.

Proof. Residual SVD optimality and the triangle inequality give $\|(I - Q_k)A_{(k)}\|_{op} \le d_k$ on the common event, even for selected r_k. The definition of R_k gives $(I - Q_k)(R_k)_{(k)} = (R_k)_{(k)}$. Operator/nuclear norm duality bounds its contraction with A by $d_k\|(R_k)_{(k)}\|_*$. Sum and combine with the norm-only bound. Conditional on the pilot, validation noise in the selected loading is Gaussian with variance $\tau_2^2\|PL\|_F^2$. No multiplicity correction over the candidate ranks is required: all bias certificates hold on one uniform pilot event, and the choice precedes validation. The union bound gives total error at most α. Unfolding SVDs, tensor projections and nuclear norms are polynomial in the explicit input size.

This algorithm never interprets an undetected component as identically zero. It does not certify the actual rank. Its bias term can be large, and no weak-signal or pointwise minimax optimality is claimed. Honesty on the full norm ball cannot yield the low-rank weak-signal minimax gain: the earlier unrestricted-class lower bound still applies. Independent artificial Gaussian splitting of one observation is valid with $\tau_1 = \tau_2 = \sqrt{2}\tau$; using both halves at scale τ is invalid.

G.6.1 Why the first-order repair does not solve the original problem

For the A2 construction, if $\min_k \eta_k \ge z_{\mathrm{val}}\tau_2$, where $z_{\mathrm{val}} = z_{1-(\alpha-\delta)/2}$, no projected candidate improves the smaller of the untruncated direct and norm-only radii. In fact, nonzero R_k have $d_k \ge \eta_k$, so triangle inequalities and nuclear/Frobenius norm comparison give $B \ge \min(M, \min_k \eta_k)\|R\|_F$. Hence its radius is at least $\min(M, z_{\mathrm{val}}\tau_2)(\|PL\|_F + \|R\|_F)$, which is at least $\min(M, z_{\mathrm{val}}\tau_2)\|L\|_F$. The explicit envelopes meet this condition in the equal-noise experiments below. This is a proved limitation of this particular repair, not a universal computational lower bound.

G.7 A3. The adaptation obstruction at an arbitrary orientation

Let a base tensor A_0 have norm $a < M$, and leave at least one unused unit of every supplied Tucker or TT rank cap. Choose mode subspaces W_k such that A_0 is orthogonal to every tensor in $\mathcal{W} = \bigotimes_k W_k$. One sufficient choice is to take at least one W_k orthogonal to the column space of $A_{0,(k)}$. Put $M_0 = \sqrt{M^2 - a^2}$ and $L_{\mathcal{W}} = P_{\mathcal{W}}L$. Then

$$\mathbb{E}_{A_0}|C| \ge c_\alpha \min(M_0, \tau)\rho(L_{\mathcal{W}})$$

for every interval honest on the outer class. If the restriction of the loading to an m-dimensional orthogonal diagonal block in $\mathcal{W}$ equals b times a flat unit loading, the stronger bound is

$$\mathbb{E}_{A_0}|C| \ge c_{K,\alpha}|b|\,\mathfrak{r}_K(m, M_0, \tau).$$

Here ρ is taken within the indicated product subspace. The zero overlap case gives a zero lower bound and says nothing further.

Proof. Add rank-one perturbations $B \in \mathcal{W}$ with norm at most M_0. Rank subadditivity keeps $A_0 + B$ in every outer unfolding cap; orthogonality gives $\|A_0 + B\|_F^2 = a^2 + \|B\|_F^2 \le M^2$. Translate the Gaussian experiment by A_0. The target difference is $\langle L_{\mathcal{W}}, B\rangle$. Apply the two-point construction of L1 or the existing flat-loading mixtures, all of which have rank one. Unused observed coordinates contain identical independent Gaussian noise under both hypotheses and do not change the likelihood

calculation. This is a rotation-invariant extension of the preceding block-supported shifted bound, not a complete characterization at every mean.

G.8 A4. A cap-valid tangent interval on the original unseparated class

There is a more useful repair when mode caps $\bar{r}_k \le p_k$ are supplied. Use the same independent Gaussian pilot and validation data and envelopes as A2. Let Q_k project onto the leading $\bar{r}_k$ pilot left singular vectors. Define

$$e_k = \begin{cases} 0, & \bar{r}_k = p_k, \\ \min\{1, 2\eta_k/(\sigma_{\bar{r}_k}(Z^{(1)}_{(k)}) - \eta_k)\}, & \bar{r}_k < p_k,\ \sigma_{\bar{r}_k}(Z^{(1)}_{(k)}) > \eta_k, \\ 1, & \text{otherwise.} \end{cases}$$

With the first-order product-space projector from the previous stage,

$$T = \bigotimes_k Q_k + \sum_k (I - Q_k) \otimes \bigotimes_{l \ne k} Q_l,$$

put $R = (I - T)L$ and

$$B_T = M\|R\|_F \min\{1, (\sum_{k<l} e_k^2 e_l^2)^{1/2}\}.$$

The interval centered at $\langle TL, Z^{(2)}\rangle$, with radius $B_T + z_{\mathrm{val}}\tau_2\|TL\|_F$, is honest on **every** tensor with these supplied mode caps and norm bound, including zero, arbitrarily weak signals, and actual ranks strictly below the caps. No singular-value separation is assumed. Choose the smallest pilot-measurable radius among this candidate, the direct interval, and the structural norm interval, as before.

Proof. Let V_k be the actual left singular subspace of $A_{(k)}$. On the simultaneous noise event we claim $\|(I - Q_k)P_{V_k}\|_{op} \le e_k$. This is immediate for $e_k = 1$ or $Q_k = I$. Otherwise $\sigma_{\bar{r}_k}(Z^{(1)}_{(k)}) > \eta_k$ forces the true mode rank to equal the supplied cap: a smaller true rank would give that observed singular value at most η_k by Weyl's inequality. The pilot tail after the cap is at most η_k, so $\|(I - Q_k)A_{(k)}\|_{op} \le 2\eta_k$. Multiplying by the inverse nonzero singular values and using $\sigma_{\bar{r}_k}(A_{(k)}) \ge \sigma_{\bar{r}_k}(Z^{(1)}_{(k)}) - \eta_k$ proves the claim.

Decompose A into the mutually orthogonal mode-projector blocks indexed by which modes fall in $I - Q_k$. The complement of T consists of blocks with at least two such modes. Counting each block once is bounded by counting it for every contained pair. Consequently

$$\|(I - T)A\|_F^2 \le \sum_{k<l} \|(I - Q_k)(I - Q_l)A\|_F^2 \le M^2 \sum_{k<l} e_k^2 e_l^2.$$

The last inequality restricts each operator to its true mode subspace and uses that operators on distinct modes commute. Combine with the trivial bound $\|(I - T)A\|_F \le M$ and Cauchy–Schwarz against R. Conditional Gaussian validation and the common pilot event now prove coverage and valid radius selection exactly as in A2.

This algorithm is polynomial and honest for the original weak-signal class. Strong signals can make the second-order bias correction small, even though separation is not a coverage assumption. It is **not** a polynomial weak-signal minimax solution or a method of estimating actual rank. At smaller actual ranks, the unresolved cap singular values trigger conservative error bounds. For a supplied TT cap vector, valid mode caps are $\min(p_k, \bar{r}_{k-1}\bar{r}_k)$, with boundary TT ranks one; the interval does not estimate TT bond ranks. Artificial splitting must use its increased noise scales.

For the original single observation, `single_observation_interval` creates $Z \pm \tau G'$ with independent artificial G', uses scales $\sqrt{2}\tau$ and error $\alpha/2$ for the cap-valid construction, and intersects with the direct interval based on Z at error $\alpha/2$. The union bound proves honesty at error α despite the dependence

between these two intervals. The direct component caps the reported length at $2z_{1-\alpha/4}\tau\|L\|_F$. The split is randomization within the original experiment, not an extra observed replicate.

G.9 Remaining problems

1. Polynomial minimax intervals for diffuse weak-signal classes at general dimensions: open. The new optimizer establishes a working global benchmark only for binary rank-one tensors; it does not remove the computational obstruction.
2. Sharp larger-budget rates for every loading, including every unequal weight sequence: open. L1 resolves all loadings only for $M \le \tau$; L2 strengthens the usable lower-bound machinery.
3. General rank-dependent and pointwise adaptation: open, with explicit impossibility results for some oracle demands. Rate adaptation over fixed positive nested caps is already attained statistically (A1).
4. General Tucker/TT global optimizers, validated directed-rounding arithmetic, and high-dimensional runtime scaling: not implemented.

H Growing caps, decaying loadings and pointwise inference

Observe $Z = A + \tau G$ with independent standard Gaussian tensor entries, known $\tau > 0$, norm bound $\|A\|_F \le M$, and supplied Tucker or TT caps. Unless specified otherwise, dimensions are equal to p and $K \ge 3$. For $L = p^{-1/2}\sum_{i=1}^{p} E_i$, write $\psi(A) = \langle L, A\rangle$, where $E_i = e_i^{\otimes K}$. The criteria $\mathcal{L}_0$ and $\mathcal{L}_{\max}$ are optimal expected length at zero and optimal worst-case expected length among intervals honest over the whole stated class. Fix $0 < \alpha < 1/4$.

H.1 G1. Sharp growing-cap rates with a polynomial algorithm

Let $r = \min(p, \min_k \bar{r}_k)$, where the minimum is over the supplied constrained unfoldings (Tucker modes or TT cuts). Suppose $r \ge c\sqrt{p}$ for a fixed $c > 0$. Uniformly over all remaining caps, all p, and every $M, \tau > 0$,

$$\mathcal{L}_0 \asymp_{\alpha,c} \mathcal{L}_{\max} \asymp_{\alpha,c} \min\{M\sqrt{r/p}, \tau\}.$$

Minimax squared risk has the square of this rate. The constants do not depend on growing ranks or tensor order. Choose deterministically between the structural interval $[-h_0, h_0]$, $h_0 = M\sqrt{r/p}$, and the direct Gaussian interval centered at $\psi(Z)$ with radius $z_{1-\alpha/2}\tau$. This takes $O(p)$ accessed entries and arithmetic for the coordinate loading. For a rotated flat loading, obtaining the contractions has its additional polynomial input-access cost. No factor estimation or exponential scan is required.

Upper proof. An unfolding attaining the smallest cap has loading matrix singular values $1/\sqrt{p}$, repeated p times. Matrix singular-value duality and Cauchy–Schwarz bound its contraction with any rank-at-most-r matrix of norm M by $M\sqrt{r/p}$. Thus the structural interval covers surely. The direct error is $N(0, \tau^2)$. Deterministic radius selection proves the length upper bound. Choosing estimator zero when $h_0 \le \tau$ and the direct estimator otherwise gives squared risk at most $\min(h_0^2, \tau^2)$.

Lower proof. For a uniform size-r subset S of $\{1, \dots, p\}$ set

$$A_S = b\sum_{i\in S} E_i, \qquad b = \min\{M/\sqrt{r}, \tau\sqrt{\log(1 + p/(4r^2))}\}.$$

All constrained unfolding ranks are r, the norm is at most M, and the common target is $\Delta = rb/\sqrt{p}$. For independent supports S, T, the full Gaussian likelihood second moment equals $\mathbb{E}\exp\{b^2|S\cap T|/\tau^2\}$. To bound it, for $H = |S \cap T|$ factorial moments give $\mathbb{E}\binom{H}{j} = \binom{r}{j}(r)_j/(p)_j \le \binom{r}{j}(r/p)^j$. Expand $(1+t)^H$ at $t = e^{b^2/\tau^2} - 1$ to obtain

$$1 + \chi^2(Q, P_0) \le (1 + rt/p)^r \le \exp(r^2 t/p) \le e^{1/4}.$$

Put $x = p/(4r^2) \le 1/(4c^2)$. Since $x/(1+x) \le \log(1+x) \le x$,

$$\frac{\tau}{2\sqrt{1 + 1/(4c^2)}} \le \frac{\tau r}{\sqrt{p}} \sqrt{\log(1 + p/(4r^2))} \le \frac{\tau}{2}.$$

Consequently $\Delta \asymp_c \min(M\sqrt{r/p}, \tau)$. Total variation is at most $\eta = \sqrt{e^{1/4} - 1}/2 < 1/2$. Coverage of both zero and the common target implies $\mathbb{E}_0|C| \ge \Delta(1 - 2\alpha - \eta)$, which is positive for $\alpha < 1/4$. The nearest-target testing argument also gives squared risk at least $\Delta^2(1 - \eta)/8$. This proves all claims.

This closes the large-cap regime using the earlier sparse-diagonal lower bound. It is a consequence of that bound with a newly matched upper bound, not a new sparse-mixture technique. Under $M = \tau\sqrt{p}$ the rate is τ, so the vanishing fixed-cap improvement has already disappeared by rank order $\sqrt{p}$, well below unrestricted rank p.

H.2 G2. General adaptation across positive growing caps is impossible

Set $M = \tau\sqrt{p}$ and let the outer smallest cap satisfy $r \ge c\sqrt{p}$. Every interval honest on that outer class has $\mathbb{E}_0|C| \ge c_{\alpha,c}\tau$. For a fixed positive inner cap vector, the previously established oracle minimax rate is

$$\mathfrak{r}_K(p) = \frac{\tau}{\sqrt{p}} \begin{cases} (\log(ep))^{3/2}, & K = 3, \\ \sqrt{\log(ep)}, & K \ge 4. \end{cases}$$

Zero belongs to every such inner class. Its worst-case expected length under outer honesty is therefore at least order τ, a diverging factor $\sqrt{p}/(\log(ep))^{3/2}$ or $\sqrt{p/\log(ep)}$ larger than its known-class oracle rate. No algorithm, including an exponential one, can be simultaneously minimax in rate over these positive fixed and growing cap classes. This is stronger than excluding only the known-rank-zero oracle. It does not conflict with earlier adaptation across *fixed* outer caps, whose constants were allowed to depend on that outer cap.

Nonzero version. Suppose A_0 is supported on the first q coordinates in every mode, with norm $a < M$ and actual constrained ranks $r_{0,k}$. Let $m = p - q$, $s = \min(m, \min_k(\bar{r}_k - r_{0,k}))$ and $M_0 = \sqrt{M^2 - a^2}$. If $s \ge c\sqrt{m}$, then

$$\mathbb{E}_{A_0}|C| \gtrsim_{\alpha,c} \min\{M_0\sqrt{s/p}, \tau\sqrt{m/p}\}.$$

Proof. Add the G1 alternatives on the disjoint remaining block. Norm squares add; every unfolding rank increases by at most s. Translation by A_0 preserves Gaussian likelihood overlaps, and normalization of the target multiplies the dimension-m lower bound by $\sqrt{m/p}$. This gives a positive nonzero-base obstruction when weak extra rank fits inside the supplied caps. Arbitrarily oriented versions follow by using orthogonal product blocks when the transformed loading is flat on that block.

H.3 W1. A dyadic loading bound using the full weight profile

Return to a prescribed orthogonal product loading $L_w = \sum_{i=1}^m w_i U_i$, with decreasing absolute weights and fixed positive caps. The ambient mode dimensions may exceed m. Partition the ordered indices into $B_0 = \{1\}$ and $B_j = \{2^{j-1} + 1, \dots, \min(2^j, m)\}$ for $j \ge 1$, stopping when empty. Let $a_j = \max_{i \in B_j} |w_i|$, $d_j = |B_j|$, and $\alpha_j = \alpha/2^{j+1}$. Define

$$H_K(w) = \sum_j a_j (j+1)^{\nu_K}, \qquad \nu_3 = 3/2, \quad \nu_K = 1/2 \ (K \ge 4).$$

There is a full-data honest interval satisfying the deterministic bound

$$|C_w| \le C_{K,\bar r,\alpha} \min\{M\|w\|_\infty, \tau H_K(w)\}.$$

The minimax squared risk is at most a constant times the square of the same expression. This is a statistical upper bound using the ideal block residual fits, not a polynomial implementation of those fits.

Proof. Rotate the known orthogonal factors into coordinates. Restrict the data to each d_j^K block; this cannot increase its norm or rank caps. Apply the earlier weighted residual confidence projection at error α_j. With $J_2 = J(2\bar r)$, its block diameter is at most

$$4J_2^{K/2} a_j \sum_{s=1}^{d_j} \lambda_s^{(j)} s^{-K/2},$$

where

$$\lambda_s^{(j)} = \tau\sqrt{2\{s\log(ed_j/s) + Ks\log(1+4K) + \log(2d_j/\alpha_j)\}}.$$

For $s \ge 1$, $\log(ed_j)$ and $\log(1/\alpha_j)$ are bounded by a fixed constant times $j+1$. Thus $\lambda_s^{(j)} \le C_{K,\alpha}\tau\sqrt{s(j+1)}$. The remaining sum is harmonic for $K=3$, bounded by $C(j+1)$, and bounded by a K-dependent constant for $K \ge 4$. Add the block intervals and use $\sum_j \alpha_j \le \alpha$. Their lengths sum to at most $C\tau H_K(w)$. The old structural radius is at most $M\sqrt{\min \bar r_k}\|w\|_\infty$, which is $O(M\|w\|_\infty)$ for fixed caps. Select the smaller deterministic bound before seeing the data.

For squared error use block residual fits with error budgets $\beta_j = 2^{-j-4}$ and deterministic optimization tolerance τ. The earlier excess-tail proof gives each block error an L_2 norm at most $C\tau a_j(j+1)^{\nu_K}$: the deterministic diameter term has the bound just derived, and the excess term has L_2 norm $O(\tau a_j)$ because $\sum_s s^{-K/2} < \infty$ and $\mathbb{E}T_{0,j}^2 \le 2\beta_j\tau^2$. Minkowski's inequality bounds the L_2 norm of their summed error by $C\tau H_K(w)$, without requiring independence. Compare with estimator zero and its structural risk bound. The finite number of blocks makes measurability and summation immediate.

H.4 W2. Sharp rates at every budget for polynomially decreasing loadings

Suppose for fixed $\beta > 0$ and $C_0 < \infty$ that the nonzero, decreasing weights satisfy

$$|w_i| \le C_0|w_1|i^{-\beta}, \qquad i = 1,\dots,m.$$

For fixed positive caps and order, uniformly over all $m, M, \tau > 0$,

$$\mathcal{L}_0(\Theta; L_w) \asymp \mathcal{L}_{\max}(\Theta; L_w) \asymp |w_1|\min(M,\tau),$$

and minimax squared risk is $\asymp |w_1|^2\min(M,\tau)^2$. Constants may depend on $\beta, C_0, K, \bar r, \alpha$, but not on dimension or budget. This includes all signed power-law weights, whether normalized or not. In particular, $0 < \beta < 1/2$ permits diffuse unit-Frobenius loadings with $|w_1| \to 0$; the result is not limited to a constant product overlap with a unit-Frobenius loading.

Proof. For $j \ge 1$, $a_j \le C_0|w_1|2^{-\beta(j-1)}$. The series $\sum_j 2^{-\beta(j-1)}(j+1)^{\nu_K}$ is finite for every fixed $\beta > 0$. W1 therefore gives the matching length and risk upper bounds. For the lower bounds compare zero with tU_1, where $t = \min(M,\tau)/2$ and its sign agrees with w_1. These rank-one means belong to every positive cap class, their target difference is $t|w_1|$, and their total variation is at most $2\Phi(1/4) - 1 < 1/4$. The common-target interval argument and two-point squared-loss argument give the asserted lower orders. No lower power-envelope assumption on the other weights is needed.

As $\beta \downarrow 0$ the constants deteriorate: the dyadic sum is of order at most $1 + \beta^{-(\nu_K+1)}$ up to fixed constants. Thus this theorem does not cover a dimension-dependent exponent approaching zero uniformly,

and does not contradict the extra logarithmic factors for flat loadings. Completely arbitrary unequal weight sequences, including many slowly varying profiles, remain uncharacterized.

Nonorthogonal perturbations. Suppose $L = L_w + R$, with the same weight condition and a known otherwise arbitrary tensor R satisfying $\|R\|_F \le \epsilon|w_1|$ for a fixed $0 \le \epsilon < 1$. The same sharp rates hold with constants depending also on $1-\epsilon$. For the lower bound, $|\langle L, U_1\rangle| \ge (1-\epsilon)|w_1|$. For large budgets add a direct interval for $\langle R, A\rangle$ at a separate error allocation, costing $O(\tau\|R\|_F)$. For small budgets, $\rho(L_w) = |w_1|$ by generalized Hölder and $\rho(R) \le \|R\|_F$; the canonical expansion bounds the structural radius by $\sqrt{J}M(1+\epsilon)|w_1|$. The same reasoning applies to squared error, using the direct residual estimator and Minkowski. This gives an all-budget result for a class of loadings without an exact orthogonal representation, but is not a theorem for every loading.

H.5 W3. A restricted computational separation for these newly matched loadings

The W2 rate does not automatically have an efficient estimator. To make this distinction precise, let $w_i = w_1 i^{-\beta} > 0$ with fixed $0 < \beta < 1/K$, and set $M = \tau\sqrt{p}$. Choose constants

$$\frac{1}{K} < a < \min\left\{\frac{1}{2}, \frac{1-2\beta}{K-2}\right\}, \qquad 0 < b < a - \frac{1}{K}.$$

For $s = \lfloor p^a \rfloor$ and $\mathsf{d} = \lfloor p^b \rfloor$, every polynomial estimator of total degree at most d in the full observed tensor has worst-case squared risk polynomially larger than the unrestricted $\tau^2 w_1^2$ rate of W2.

Finite statement and proof. For a uniform size-s support S, put $x_S = s^{-1/2}\mathbf{1}_S$ and let d_w be the diagonal map with entries $w_p/w_i \le 1$. Use the rank-one alternatives

$$A_S = \tau\lambda(\mathsf{d}_w x_S)^{\otimes(K-1)}, \qquad 0 < \tau\lambda \le M.$$

They have norm at most M and common weighted target $\Delta = \tau\lambda w_p s^{1-K/2}$. All factors are nonnegative, so for two supports $0 \le \langle A_S, A_T\rangle/\tau^2 \le \lambda^2|S\cap T|^K/s^K$. The full Gaussian Hermite expansion therefore bounds the squared norm of the centered likelihood ratio through degree d by

$$\chi^2_{\le \mathsf{d}} \le \sum_{d=1}^{\mathsf{d}} \frac{\lambda^{2d}\mathbb{E}H^{Kd}}{d!\,s^{Kd}} \le \frac{\mu e^{\mu} B_{\mathsf{d}}}{1-B_{\mathsf{d}}} = \eta^2_{\mathsf{d}}, \quad \mu = s^2/p, \quad B_{\mathsf{d}} = \frac{\lambda^2(K\mathsf{d})^K}{s^K} < 1.$$

For completeness, counting maps by their nonempty fibers gives $H^q = \sum_{j=1}^q S(q,j)(H)_j$, where $S(q,j)$ counts partitions of q elements into j nonempty sets. We have $S(q,j) \le j^q/j!$ and $\mathbb{E}(H)_j \le \mu^j$. Hence $\mathbb{E}H^q \le q^q(e^{\mu}-1) \le \mu e^{\mu} q^q$. Substitute $q = Kd$ and sum the geometric upper bound to get the display. For a degree-d estimator with worst-case squared risk R, orthogonality of Hermite polynomials and Cauchy–Schwarz give $|\mathbb{E}_Q T - \mathbb{E}_0 T| \le \sqrt{R}\,\eta_{\mathsf{d}}$. The two target means differ by Δ and their estimation biases are each at most $\sqrt{R}$, so

$$R \ge \Delta^2/(2+\eta_{\mathsf{d}})^2.$$

Set $\lambda = \sqrt{p}$. Then $\mu \to 0$ and $B_{\mathsf{d}} = O(p^{1-Ka+Kb}) \to 0$, whereas

$$\frac{\Delta}{\tau w_1} \asymp p^{1/2-\beta-a(K-2)/2} \longrightarrow \infty$$

at a polynomial rate by the choice of a. Such an a exists precisely under the stated sufficient condition $\beta < 1/K$. This proves the restricted separation, using the earlier general low-degree argument with a new loading-compatible prior. It includes diffuse power-law loadings with sharp all-budget rates from W2.

This is not a lower bound for all polynomial-time algorithms or all confidence intervals. An interval honest on zero and these alternatives would give the test $\mathbf{1}\{\Delta \in C(Z)\}$ with total error at most

$2\alpha + \mathbb{E}_0|C|/\Delta$. Excluding such efficient tests requires an additional computational hypothesis; the polynomial-estimator result alone does not supply one.

H.6 P1. Efficient pointwise optimality at strong saturated Tucker means

This result concerns **Tucker mode caps all equal to** r, $1 \le r \le p$. It does not substitute TT bond caps for mode caps. Let

$$A_0 = \sum_{i \in S} a_i E_i, \qquad |S| = r, \quad a = \min_{i \in S} |a_i| > 0, \quad \|A_0\|_F + \tau \le M \le c_0 a \sqrt{r},$$

where $c_0 > 1$ is fixed. The support and signed amplitudes are not supplied to the algorithm. These means saturate every supplied mode cap. There is a polynomial interval honest on the entire unseparated Tucker class such that, in the strong-signal regime specified below,

$$\mathbb{E}_{A_0}|C| \asymp_{K,\alpha,c_0} \tau\sqrt{r/p},$$

and this matches the infimum expected length at A_0 under that *same outer-class honesty requirement*. Both p and r may grow. This is pointwise optimality in rate, not optimal leading constants.

Algorithm and noise budgets. Make the valid artificial split $Z^{(1)} = Z + \tau G'$, $Z^{(2)} = Z - \tau G'$, with independent errors of scale $v = \sqrt{2}\tau$. Set $\delta = \alpha r/(8p)$ and

$$\eta = 2v\sqrt{2\{(p + p^{K-1})\log 9 + \log(2K/\delta)\}}.$$

Use the preceding cap-valid tangent interval with pilot failure budget δ and total failure budget $\alpha/2$. Its validation quantile is $z_{\mathrm{val}} = z_{1-(\alpha/2-\delta)/2}$. Intersect it with the direct interval on Z at error $\alpha/2$ and the structural range. All choices are valid on the unseparated class. The total reported length is bounded by $2z_{1-\alpha/4}\tau$, including on pilot failure.

A sufficient condition for the pointwise rate is

$$a \ge 4\eta, \qquad a \ge \eta^2\sqrt{p}/\tau.$$

For fixed order/confidence this is satisfied at sufficiently large constant multiple of $\tau p^{K-1/2}$. This requirement is conservative and much stronger than a claim of optimal computational signal strength.

Upper proof. On the simultaneous pilot event, the top-r projector Q_k satisfies $\|Q_k - P_k\|_{op} \le d = 4\eta/a$, where P_k projects onto the coordinates in S. The cap-valid certificate has each angle bound at most d by Weyl's inequality and $a \ge 4\eta$. Write T_Q and T_P for their first-order product-space projectors. The representation

$$T_Q = \sum_k \bigotimes_{l \ne k} Q_l - (K-1)\bigotimes_l Q_l$$

with an identity in the omitted mode, and product telescoping, gives $\|T_Q - T_P\|_{op} \le 2K(K-1)d$. For the diagonal loading, $T_P L = p^{-1/2}\sum_{i \in S} E_i$, so

$$\|T_Q L\|_F \le \sqrt{r/p} + 2K(K-1)d.$$

The cap-valid bias is bounded by $M\sqrt{\binom{K}{2}}d^2 \le 16c_0\sqrt{\binom{K}{2}}\sqrt{r}\,\eta^2/a$. The assumed amplitude makes this $O(\tau\sqrt{r/p})$. It also gives $d \le 4\tau/(\eta\sqrt{p}) = O(\sqrt{r/p})$, because $\eta \gtrsim \tau$ and $r \ge 1$. Thus the selected/intersected interval has length $O(\tau\sqrt{r/p})$ on the pilot event. Off that event its length is $O(\tau)$, with probability at most $\delta = O(r/p) \le O(\sqrt{r/p})$. The expected-length upper bound follows. The dimension-dependent failure budget is essential to this argument; fixed failure probability would leave an unjustified $O(\tau)$ term.

Lower proof. Put $V = r^{-1/2}\sum_{i\in S} E_i$ and compare A_0 with $A_0 + (\tau/2)V$. Both have mode ranks at most r and norms at most M. Their full Gaussian total variation is less than $1/4$ and their target difference is $(\tau/2)\sqrt{r/p}$. The common-target interval argument at base A_0 proves the lower bound. This lower comparison uses the same outer class on which the algorithm is honest.

Computation. Economy SVDs of the $p \times p^{K-1}$ unfoldings require $O(Kp^{K+1})$ arithmetic and $O(p^K + p^2)$ working storage for fixed K. Tensor projection has polynomial cost of the same order. The new code uses economy SVDs; the previous prototype requested full right singular matrices, which could allocate p^{2K-2} entries. Thus the implementation now scales with the explicit tensor rather than that avoidable allocation. It still reads the full tensor and does not claim sublinear access cost.

H.7 What these results do and do not resolve

- Efficient high-dimensional minimax computation is resolved here for $r \gtrsim \sqrt{p}$, where G1 gives the sharp all-budget rate. Original fixed-cap diffuse-loading weak signals remain computationally open.
- All-budget loading rates are resolved for fixed polynomial decay and the specified nonorthogonal perturbations. General profiles remain open; the matching W2 statistical construction still uses ideal block fits.
- General growing-cap adaptation is impossible in the G2 comparison, even between positive cap classes. P1 gives a positive, efficiently attained pointwise rate on a strong saturated Tucker family. It does not characterize pointwise lengths everywhere or optimal SNR thresholds.

No blanket tensor hardness theorem is inferred from low-degree results for sparse tensor PCA. Those results concern a restricted computational model; a testing reduction for arbitrary efficient intervals still needs an explicit, justified computational hypothesis.

H.8 Methodological references

- The sparse-functional precursor for G1 is Collier, Comminges and Tsybakov, *Minimax estimation of linear and quadratic functionals on sparsity classes*.
- For the distinct estimation-adaptation problem, see Collier, Comminges, Tsybakov and Verzélen, *Optimal adaptive estimation of linear functionals under sparsity*.
- The restricted computational perspective in W3 follows the methodology of Choo and d'Orsi, *The Complexity of Sparse Tensor PCA*.
- For general tensor-functional tangent inference under signal conditions, see Xu, Chen and Han, *Statistical Inference for Low-Rank Tensor Models*.

The repository-specific dependency and novelty distinctions are recorded in the accompanying audit and literature update.

I R1. Scalar companion and risk proof

This is a new statement for the revision, with the complete argument below. It does not alter the confidence interval or assert pointwise minimax squared risk.

Let $Y = \psi + \tau g$, $g \sim N(0,1)$, and suppose the supplied structural range is $|\psi| \le h$. Let I be the intersection of any candidate interval with $[Y - d, Y + d]$ and $[-h, h]$, where $d = z_{1-\alpha/4}\tau$. The candidate may depend arbitrarily on the data. The reported confidence interval may retain its existing empty-intersection convention $\{0\}$.

Define a **separate scalar estimator** by $\widehat{\psi} = 0$ when $h \le \tau$. Otherwise, use the midpoint of the actual nonempty intersection; on an empty intersection use $\Pi_{[-h,h]}Y$. Replacing an exactly zero singleton by the clipped direct estimate also satisfies the bound. Then, for every admissible mean,

$$E(\widehat{\psi}-\psi)^2 \le \begin{cases} h^2, & h \le \tau, \\ \tau^2\{1 + 2z_{1-\alpha/4}\sqrt{2/\pi} + z_{1-\alpha/4}^2\}, & h > \tau. \end{cases}$$

Thus the risk is at most $C_\alpha \min(h^2, \tau^2)$, uniformly over the norm budget and signal magnitude. No candidate coverage, separation, rank recovery, or independence from Y is needed for this **risk** bound. Honesty of the reported interval continues to require its original assumptions and calibration.

Proof. On $h \le \tau$, zero has squared error at most h^2. Otherwise, on a nonempty intersection its midpoint m is in $[Y-d, Y+d]$, so $|m-\psi| \le |Y-\psi| + d$. On an empty intersection, projection onto an interval containing ψ is nonexpansive relative to ψ: $|\Pi_{[-h,h]}Y - \psi| \le |Y-\psi|$. The same argument covers a zero-singleton replacement. Consequently the error on every observation is at most $\tau|g| + d$. Taking expectations, using $Eg^2 = 1$ and $E|g| = \sqrt{2/\pi}$, proves the second branch. Since $C_\alpha \ge 1$, both branches imply the final inequality. No conditioning on a successful pilot or deletion of rare events occurs.

For A4/P1 and the normalized diagonal loading, $h = M\sqrt{r/p}$. The cap-valid intervals already intersect the direct interval at error $\alpha/2$, so $d = z_{1-\alpha/4}\tau$ is exactly their existing direct radius. The scalar companion therefore removes the M^2 loss from the arbitrary zero sentinel without changing any coverage event or interval length. For general fixed L, replace τ by $\tau\|L\|_F$ and use any valid supplied structural range. This is a noise-scale risk bound, not a proof of the smaller P1 pointwise squared-risk rate or of the ideal full-data minimax risk.

References

T. Tony Cai and Mark G. Low. Minimax estimation of linear functionals over nonconvex parameter spaces. *The Annals of Statistics*, 32(2):552–576, 2004. doi: 10.1214/009053604000000094. URL https://arxiv.org/abs/math/0406427.

Davin Choo and Tommaso d'Orsi. The complexity of sparse tensor PCA. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://arxiv.org/abs/2106.06308.

Olivier Collier, Laëtitia Comminges, and Alexandre B. Tsybakov. Minimax estimation of linear and quadratic functionals on sparsity classes. *The Annals of Statistics*, 45(3):923–958, 2017. URL https://arxiv.org/abs/1502.00665.

Luc Devroye, Matthieu Lerasle, Gábor Lugosi, and Roberto I. Oliveira. Sub-gaussian mean estimators. *The Annals of Statistics*, 44(6):2695–2725, 2016. doi: 10.1214/16-AOS1440.

Daniel Fu and Youngtak Sohn. Low-degree estimation thresholds in planted hypergraphs and tensor PCA, 2026. URL https://arxiv.org/abs/2605.30113.

Philip B. Stark. Affine minimax confidence intervals for a bounded normal mean. *Statistics & Probability Letters*, 13(1):39–44, 1992. doi: 10.1016/0167-7152(92)90234-V.

Ke Xu, Elynn Chen, and Yuefeng Han. Statistical Inference for Low-Rank Tensor Models, 2025. URL https://arxiv.org/abs/2501.16223.